1 Recent Progress in Research on the Pharmacological Potential of Mushrooms and Prospects for Their Clinical Application

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Abstract

Fungi are considered one of the most diverse, ecologically significant, and economically important organisms on Earth. The edible and medicinal mushrooms have long been known by humans and were used by ancient civilizations not only as valuable food but also as medicines. Mushrooms are producers of high- and low-molecular-weight bioactive compounds (alkaloids, lectins, lipids, peptidoglycans, phenolics, polyketides, polysaccharides, proteins, polysaccharide-protein/peptides, ribosomal and non-ribosomal peptides, steroids, terpenoids, etc.) possessing more than 130 different therapeutic effects (analgesic, antibacterial, antifungal, anti-inflammatory, antioxidant, antiplatelet, antiviral, cytotoxic, hepatoprotective, hypocholesterolemic, hypoglycemic, hypotensive, immunomodulatory, immunosuppressive, mitogenic/regenerative, etc.).

The early record of *Materia Medica* shows evidence of using mushrooms for treatment of different diseases. Mushrooms were widely used in the traditional medicine of many countries around the world and became great resources for modern clinical and pharmacological research. However, the medicinal and biotechnological potential of mushrooms has not been fully investigated. This review discusses recent advances in research on the pharmacological potential of mushrooms and perspectives for their clinical application.

Keywords

Bioactive compounds · Clinical application · Ethno-mycopharmacology · Medicinal mushrooms · Pharmacological potential

Abbreviations

ACE	Angiotensin-converting enzyme
AIDS	Acquired immune deficiency syndrome
BDNF	Brain-derived neurotrophic factor
CL	Cultural liquid
COX-1	Cyclooxygenase-1
COX-2	Cyclooxygenase-2
CSF	Colony-stimulating factor
CVD	Cardiovascular diseases
DENV-2	Dengue virus type 2
DS	Dietary supplement
EPS	Exopolysaccharides
FIP	Fungal immunomodulatory protein
GLPS	*G. lucidum* polysaccharide

GSK-3	Glycogen synthase kinase 3
HIV	Human immunodeficiency virus
HK2	Hexokinase 2
HMG-CoA	5-Hydroxy-3-methylglutaryl-coenzyme A
HPV-1	Human papillomavirus 1
HSV-2	Herpes simplex virus 2
IC_{50}	The half maximal inhibitory concentration
IFN	Interferon
IL	Interleukin
iNOS	Inducible NO synthase
LPS	Lipopolysaccharide
MDCK	Madin-Darby canine kidney cells
MIC	Minimal inhibitory concentrations
MMDD	Medicinal mushroom-derived drug
MS	Mycosterol
NGF	Nerve growth factor
NO	Nitric oxide
NSAID	Nonsteroidal anti-inflammatory drug
OLTT	Oxygenated lanostane-type triterpenoid
PAMP	Pathogen-associated molecular pattern
PPAR	Peroxisome proliferator-activated receptor
PRR	Pattern recognition receptor
PI3K/Akt	Phosphatidylinositol-3-kinase and protein kinase B
PSK	Polysaccharide K
PSP	Polysaccharide-protein
PSPC	Polysaccharide-protein complex
PTP1B	Protein tyrosine phosphatase 1B
QoL	Quality of life
STAT3	Signal transducer and activator of transcription 3
TCM	Traditional Chinese medicine
TNBC	Triple-negative breast cancer
TNF	Tumor necrosis factor
TNFα	Tumor necrosis factor alfa
VDM	Vitamin D-enriched mushroom

1.1 Introduction

Mushrooms have widely been appreciated all over the world for their nutritional and medicinal properties (Moore and Chiu 2001; Chang and Miles 2008; Barros et al. 2008; Wasser 2010, 2011; Bandara et al. 2015, 2017; Valverde et al. 2015; de Mattos-Shipley et al. 2016; Chang and Wasser 2017; Gupta et al. 2018). The early

civilizations of Greek, Egyptian, Roman, Japanese, and Mexican people prized mushrooms for their therapeutic value (Hobbs 1995; Guzmán 2015). Mushrooms have been used in traditional Chinese medicine (TCM) for more than 3000 years for prevention and treatment of different diseases. There are many sources of ethno-mycopharmacological information on wild mushrooms in Eastern and Western countries, but their medicinal and biotechnological potential has not been fully exploited yet (Boa 2004; Chang and Miles 2008; Lindequist 2011; Grienke et al. 2014; Money 2016; Karun and Sridhar 2017).

The ethno-mycological and ethnographic evidence has documented that mushrooms, particularly bracket species (e.g., *Coriolopsis gallica*, *Daedalea quercina*, *Daldinia concentrica*, *Fomes fomentarius*, *Ganoderma adspersum*, *Lenzites warnieri*, *Piptoporus betulinus*, and *Skeletocutis nivea*), could be used by humans not only as food and medicine but also as tinder for fire-lighting (Roussel et al. 2002; Boa 2004; Uzunov and Stoyneva-Gärtner 2015; Berihuete-Azorín et al. 2018).

Mushrooms or macrofungi are widely distributed worldwide. They are taxonomically placed in two phyla, the *Basidiomycota* (class *Agaricomycetes*) and *Ascomycota* (class *Pezizomycetes*) in the subkingdom Dikarya. From overall numbers of fungal species estimated between 0.5-(1.5)-(5.1) million, about 140,000–160,000 species are macrofungi from which around 10% (14,000–16,000) have been identified (Hawksworth 1991, 2012; Kirk et al. 2008; Blackwell 2011; Hibbet and Taylor 2013; Tedersoo et al. 2014; Dai et al. 2015; Peay et al. 2016). From about 7000 known mushroom species, possessing various degrees of edibility, more than 3000 species from 231 genera are considered prime edible mushrooms. About 3% of the known species (at least 170, currently around 500) are poisonous, while around 700 species from 2000 known safe mushroom species have medicinal properties (Boa 2004; Barceloux 2008; Chang and Wasser 2017).

Edible/medicinal wild and cultivated mushroom species produce a broad spectrum of high- and low-molecular-weight bioactive compounds (alkaloids, lectins, lipids, peptidoglycans, phenolics, polyketides, polysaccharides, polysaccharide-proteins/peptides, proteins, ribosomal and non-ribosomal peptides, steroids, terpenoids, etc.) which possess more than 130 therapeutic effects (analgesic, antibacterial, antifungal, anti-inflammatory, antioxidant, antiviral, cytotoxic, hepatoprotective, hypocholesterolemic, hypoglycemic, hypotensive, immunomodulatory, immunosuppressive, mitogenic/regenerative, etc.) (Mizuno et al. 1995; Wasser and Weis 1999; Hawksworth 2001; Wasser 2002, 2010, 2011; Zjawiony 2004; Lindequist et al. 2005; Poucheret et al. 2006; Cerigini et al. 2007; Park et al. 2007; Stanikunaite et al. 2007; Lee et al. 2008; Saltarelli et al. 2009, 2015; Baggio et al. 2010; Ferreira et al. 2010; Xu et al. 2011; Badalyan 2012; Chang and Wasser 2012; De Silva et al. 2012a, b, 2013; Patel et al. 2012; Wang et al. 2012; Grienke et al. 2014; Park 2014; Schueffler and Anke 2014; Bandara et al. 2015, 2017; Duru and Çayan 2015; Stadler and Hoffmeister 2015; Fu et al. 2016; Kosanić et al. 2016; Venkatachalapathi and Paulsamy 2016; Chaiyasut and Sivamaruthi 2017; Chen et al. 2017; Kolundčzić et al. 2017; Kües and Badalyan 2017; Phan et al. 2017a; Sánchez 2017a, b; Shen et al. 2017; Surup et al. 2018).

The medicinal properties and therapeutic potential of mushrooms were revealed in species from different ecological groups, such as xylotrophs (e.g., *F. fomentarius*, *Fomitopsis pinicola*, *Ganoderma lucidum*, *Grifola frondosa*, *Lentinula edodes*, *Pleurotus ostreatus*, *Trametes versicolor*, *Schizophyllum commune*) and mycorrhiza-forming mushrooms (e.g., *Boletus edulis*, *Cantharellus cibarius*, *Tuber borchii*) (Chang 1996; Wasser 2010) (Figs. 1.1 and 1.2). In Asian countries, wild and cultivated mushrooms (fresh or dried) and mushroom-based bio-products are used to prevent and treat different diseases (Wasser 2010; Bandara et al. 2015), while the usage of medicinal mushrooms in Western societies is limited to

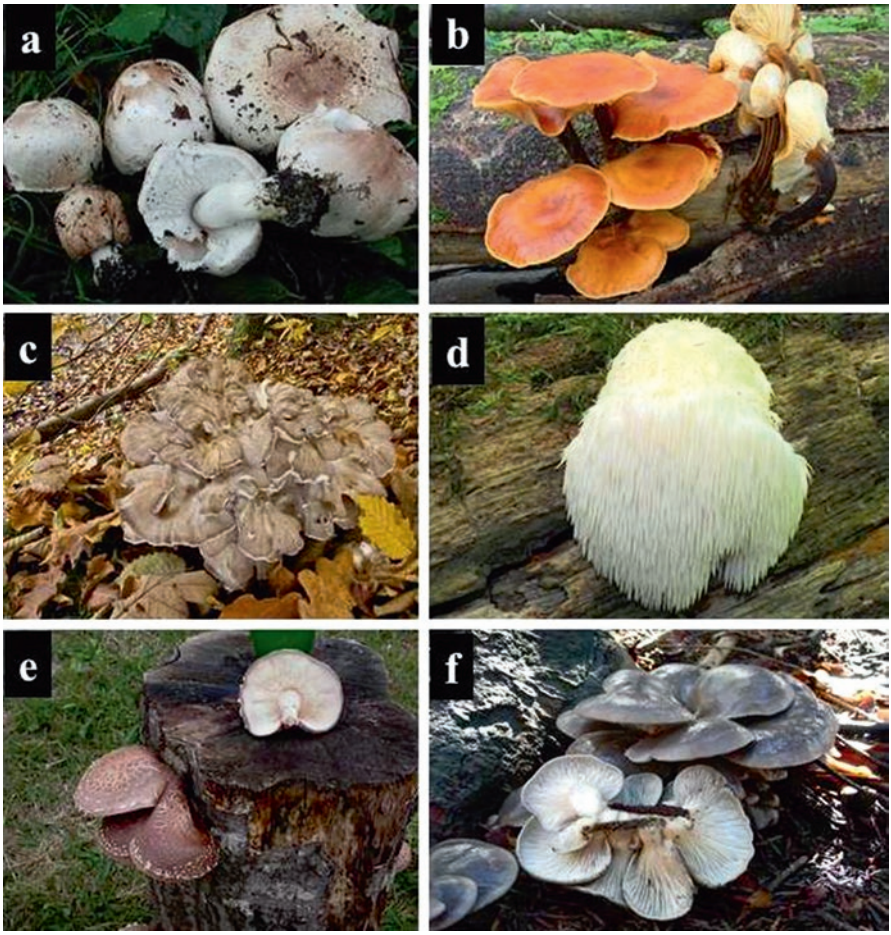

Fig. 1.1 Wild-growing fruiting bodies of edible medicinal mushrooms tested in clinical trials: (**a**) *Agaricus subrufescens* (Photo Courtesy of Guinberteau J), (**b**) *Flammulina velutipes* (Photo Courtesy of Moingeon JM), (**c**) *Grifola frondosa* (Photo Courtesy of Verstraeten P), (**d**) *Hericium erinaceus* (Photo Courtesy of Moingeon JM), (**e**) *Lentinula edodes*. (Photo Courtesy of Fourré G), (**f**) *Pleurotus ostreatus*. (Photo Courtesy of Maurice JP)

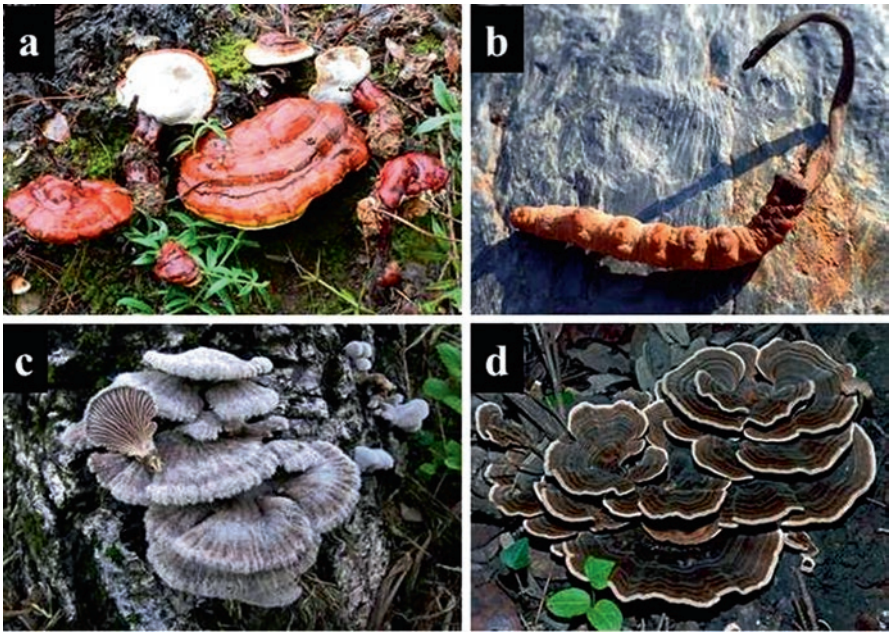

Fig. 1.2 Wild-growing fruiting bodies of inedible medicinal mushrooms tested in clinical trials: (**a**) *Ganoderma lucidum* (Photo Courtesy of Moingeon JM), (**b**) *Ophiocordyceps sinensis* (Photo Courtesy of Rioult JP), (**c**) *S. commune* (Photo Courtesy of Angelini C), (**d**) *T. versicolor*. (Photo Courtesy of Perrone L)

"nutraceuticals" (healthy dietary food) and "nutriceuticals" (functional food or DSs) (Wasser 2002, 2011; Cheung 2008; De Silva et al. 2012a, b; Chen et al. 2017). The cultivation and commercialization of natural mushroom resources will create an opportunity to study their nutraceutical and pharmacological potential for developing health-enhancing mycofood and myco-pharmaceuticals (Badalyan and Zambonelli 2019).

Mushrooms are also widely used in the production of natural cosmetic products ("cosmeceuticals" and "nutricosmetics") (Wisitrassameewong et al. 2012; Bandara et al. 2015; Taofiq et al. 2016a, c, 2017a, b; Wu et al. 2016; Badalyan and Zambonelli 2019).

The pharmacological potential of mushrooms has not been fully investigated yet. Further clinical trials are needed to substantiate the pharmacological properties or side effects of mushroom consumption before they could be recommended as health-enhancing dietary food or myco-pharmaceuticals (Money 2016; Taofiq et al. 2017a; Wasser 2017; Mustonen et al. 2018; Zmitrovich et al. 2019).

We refer our review to recent advances in the study of medicinal properties and pharmacological potential of mushrooms and prospects for their clinical application.

1.2 Biological and Genetic Resources of Medicinal Mushrooms and Their Application

The study and conservation of genetic resources and biodiversity of edible and medicinal agaricoid, polyporoid, and other groups of mushrooms, as valuable biological resources possessing high exploratory potential, including the production of nutraceuticals, nutriceuticals, pharmaceuticals, and cosmeceuticals, are carried out in different countries (Lindequist 2011; Badalyan et al. 2012, 2015; Al-Fatimi et al. 2013; Allen and Lendemer 2015; Bhattacharjee et al. 2015; Degreef et al. 2016; Badalyan and Gharibyan 2017a; Gargano et al. 2017; Badalyan and Zambonelli 2019; Bhatt et al. 2018; Gargano 2018).

Prehistoric artifacts dating back to over 5000 years ago describe the tradition of using agaricomycetous polypore fungi (order *Polyporales*) for various applications, including food and medicinal usage. Several polypores (*F. fomentarius*, *F. pinicola*, *Laetiporus sulphureus*, *Laricifomes officinalis*, and *P. betulinus*) were used in the traditional medicine of central European countries for the treatment of bladder diseases, cancer, dysmenorrhoea, hemorrhoids, pyretic diseases, and rheumatism (Lindequist et al. 2005; Lindequist 2011). Modern chemical studies have reported the presence of several primary and secondary bioactive metabolites in crude extracts of bracket fungi and a wide spectrum of their medicinal activities (anti-inflammatory, antimicrobial, cytotoxic, and others) (Suay et al. 2000; Roussel et al. 2002; Anke and Antelo 2011; Grienke et al. 2014).

Nowadays, the biological resources of edible and medicinal mushrooms are used as *dietary food* (world mushroom production was 33 million tons in 2015), *dietary supplements* (DSs) (the market for mushroom-based products is rapidly expanding and comprises more than the US $20 billion/year), *biocontrol agents* (bactericides, fungicides, herbicides, insecticides, and nematocides), *cosmeceuticals* (activator of epidermal growth factor, stimulator of collagen activity, etc.), and *mushroom-derived pharmaceuticals* or *myco-pharmaceuticals* (Wasser et al. 2000; Lindequist 2013; Kumar 2015; Gargano et al. 2017; Glamočlija and Soković 2017; Carocho et al. 2018).

1.3 Mushroom-Derived Bioactive Compounds

Fungi, including mushrooms, are considered active producers of different primary and secondary bioactive metabolites (alkaloids, fatty acids, lectins, nucleic acids, nucleosides, peptides, phenolics, polyketides, polysaccharides, proteins, statins, steroids, terpenoids, etc.) which are responsible for their pharmacological properties (antibacterial, antifungal, anti-inflammatory, antinociceptive, antioxidant, antiproliferative, antitumor, antiviral, hypocholesterolemic, hypoglycemic, hypotensive, immunomodulatory, etc.) (Badalian et al. 1996, 1997a, b, 1999, 2001; Badalian and Serrano 1999; Badalyan 2001, 2015, 2016; Badalyan and Hambardzumyan 2001;

Fang et al. 2006; Zhang et al. 2007; Wasser 2010, 2014; Lindequist 2011, 2013; Chang and Wasser 2012; De Silva et al. 2013; Khatua et al. 2013, 2017; Grienke et al. 2014; Paterson and Lima 2014; Stojković et al. 2014a, b, 2017; Duru and Çayan 2015; Valverde et al. 2015; Friedman 2016; Tao et al. 2016; Hapuarachchi et al. 2017; Khadhri et al. 2017; Masri et al. 2017; Nielsen and Nielsen 2017; Sánchez 2017a, b; Yang et al. 2017). A wide spectrum of biological activities of mushroom-derived biomolecules could be used to develop health-enhancing mycopharmaceuticals for human and animal use, such as the immunomodulatory β-glucan lentinan from *L. edodes* (Novak and Vetvicka 2008); anti-quorum-sensing agents from *Agaricus brasiliensis* (syn. *Agaricus subrufescens*) (Kerrigan 2005), as potential antibiotics (Soković et al. 2014); the antimalarial alkaloid 4-hydroxymethylquinoline from *T. versicolor* (Liu 2005); pain-suppressive enkephalinase inhibitors from *P. betulinus* (Rathee et al. 2012); and nephroprotective polysaccharides, phenolics, and flavonoids from *Pleurotus tuber-regium* (Okolo et al. 2018). The medicinal species *A. subrufescens*, *Ganoderma* spp., *G. frondosa*, *Hericium erinaceus*, *L. edodes*, *Phellinus linteus*, *P. ostreatus*, and *Polyporus umbellatus* were also reported as producers of different groups of bioactive compounds and have been recommended for a variety of therapeutic applications (Donatini 2011; Thongbai et al. 2015; Hapuarachchi et al. 2017; Thongklang et al. 2017). Among the above-mentioned species, *Ganoderma* mushrooms produce the highest diversity of bioactive compounds with different pharmacological activities (Mizuno et al. 1995; Paterson 2006; Saltarelli et al. 2009, 2015; Welti et al. 2010; Yang et al. 2013; Ma et al. 2014b, 2015; Peng et al. 2015, 2016; Klupp et al. 2015; Wang et al. 2015b, 2017a; Hapuarachchi et al. 2016a, b; Klaus et al. 2016; Meneses et al. 2016; Xu et al. 2016a; Pu et al. 2017; Stojković et al. 2017; Suárez-Arroyo et al. 2017; Subramaniam et al. 2017; Taofiq et al. 2017a; Rubel et al. 2018; Zeng et al. 2018). Alkaloids, fatty acids, nucleosides, polysaccharides, proteins, sterols, triterpenoids, and other compounds were isolated and identified from *Ganoderma* spp. They are responsible for antiaging, antibacterial, anticancer, antidiabetic, antifungal, antihypertensive, anti-inflammatory, antioxidant, antiviral, hepatoprotective, hypoglycemic, immunomodulatory, neuroprotective, wound healing, and other medicinal effects (Paterson 2006; Dai et al. 2009; De Silva et al. 2012a, b, 2013; Cheng et al. 2013; Baby et al. 2015; Bishop et al. 2015; Liu et al. 2015a; Hennicke et al. 2016; Chang et al. 2017; Chen et al. 2017; Sánchez 2017a, b; Wang et al. 2017a).

Anticancer, antimicrobial, antioxidant, antiviral, hypolipidemic, immunomodulatory, and estrogen-like activities were observed in edible oyster mushroom *Pleurotus eryngii* due to the production of diterpenoids, as eryngiolide A, hemolysins, polysaccharides, pentacyclic triterpenoids, ubiquinone-9, and other pharmacologically active biomolecules (Shibata et al. 2010; Ma et al. 2014a; Fu et al. 2016; Zhang et al. 2016; Yen et al. 2018). Recent studies of agaricomycetous species *Suillus bellinii* and *P. eryngii* demonstrated that the mycelium of *S. bellinii* possessed a higher content of ergosterol and phenolic compounds with strong antioxidant effects, while the mycelium of *P. eryngii* showed anti-inflammatory and cytotoxic effects (Souilem et al. 2017).

Genome sequencing, comparative genomics, and phylogenetic analysis of medicinal polypore mushroom *Lignosus rhinocerotis* revealed sesquiterpenoid biosynthesis genes. Moreover, the genome of *L. rhinocerotis* encodes for 1,3-β- and 1,6-β-glucans, as well as for laccase, lectin, and other fungal immunomodulatory proteins (FIP) (Yap et al. 2014, 2015).

Nowadays, pharmaceutical companies consider the medicinal mushrooms a rich source of innovative biomedical molecules extracted not only from fruiting bodies but also from cultivated mycelial biomass and cultural broth. Moreover, the mycelia and the cultural broth might be considered potential sources of bioactive compounds, due to their shorter incubation time and affordable culture conditions (e.g., requiring less space, low probability of contamination, and higher production of biomass) (Zhang et al. 2016; Bandara et al. 2017; Souilem et al. 2017). A recent review on 40 mushroom-derived pharmaceuticals available in the markets of Australia and New Zealand has demonstrated that fungal-based natural products have a major role in the future of medicine (Beekman and Barrow 2014).

1.3.1 Polysaccharides

Modern phytochemical and pharmacological studies have shown that polysaccharides are one of the major bioactive compounds in mushrooms (Wasser and Weis 1999; Zhang et al. 2007). Macrofungal β-glucans are mainly represented by β-1,3 and β-1,6 glycosidic bonds and used in the treatment of cancer because of their immunomodulatory and antitumor effects (Aleem 2013; Yoon et al. 2013; Khan et al. 2014; Piotrowski et al. 2015; Meng et al. 2016; Wang et al. 2017b). Moreover, fungal polysaccharides have less toxic side effects, unlike many existing chemotherapeutic drugs, and can be used to develop alternative medicines for supportive healthcare for the treatment of cancer (Ramberg et al. 2010). Mushroom-derived β-glucans also exhibit significant antioxidant, antiviral, and other bioactivities (Wasser and Didukh 2005; Khan et al. 2014; Kozarski et al. 2014; He et al. 2017a, b).

About 80–85% of all the existing medicinal mushroom products are (1–3)-, (1–6)-β-D-glucans, such as lentinan extracted from *L. edodes*, polysaccharide K (PSK) or Krestin derived from fruiting bodies of *T. versicolor*, schizophyllan from *S. commune*, a polysaccharide-protein complex (PSPC) or polysaccharide peptide (PSP) from *Tricholoma laboyense*, pleuran from *P. ostreatus*, as well as polysaccharides from fruiting bodies and mycelium of *H. erinaceus* (Zhu et al. 2015). The polysaccharides extracted from *Auricularia auricula-judae* possess anti-inflammatory, antioxidant, and cardioprotective effects, whereas exopolysaccharides from *P. eryngii* exhibit higher antitumor activity on human hepatoma cells (Jing et al. 2013) and suppress the proliferation of HepG-2 cells (Ma et al. 2014a).

Although there are many reports on the bioactivity and structure of fungal glucans, studies on the quantitative assessment of these compounds are scarce. The overall β-glucan content in the wild-growing mushrooms *A. auricula-judae*, *Boletus pinophilus*, *Craterellus cornucopioides*, *Gyroporus cyanescens*, *Hydnum*

repandum, *Suillus granulatus*, *S. variegatus*, and *Tricholomopsis rutilans* in comparison with edible cultivated mushroom *Agaricus bisporus* has recently been analyzed (Mirończuk-Chodakowska et al. 2017). The highest percentage of 1,3- and 1,6-β-D-glucan content in relation to the total β-glucan content was detected in *G. cyanescens* (54%), *S. granulatus* (49.8%), *A. auricula-judae* (47.9%), and *S. variegatus* (40.6%). It was shown that the average β-D-glucan content was higher in wild-growing mushrooms compared to cultivated specimens. The studied species are recommended as a source of dietary food (Mirończuk-Chodakowska et al. 2017).

The functional properties of polysaccharides obtained from medicinal mushroom *Ganoderma neo-japonicum*, as well as the presence of total phenolics, protein, and sugar in crude extracts, confirm that β-glucan-inhibiting carbohydrate-hydrolyzing enzymes may be used in the treatment of diabetes mellitus (Subramaniam et al. 2017). Biological and chemical characteristics of *Ganoderma* polysaccharides; their antioxidant, antitumor, and antimicrobial activities; as well as the structure-bioactivity relationship were recently discussed by Ferreira et al. (2015). Although *Ganoderma* polysaccharides are suggested as a healthy dietary food supplement, particularly for cancer patients, the authors considered that further clinical trials are required.

1.3.2 Terpenoids, Steroids, and Sterols

Many medicinal mushrooms contain different bioactive triterpenes, steroids, and sterols with antibacterial, antimitotic, antiviral, cytotoxic, immunomodulatory, and apoptosis-inducing effects (Baby et al. 2015; Castellano and Torrens 2015; Duru and Çayan 2015; Hadda et al. 2015a; Zhao et al. 2015, 2016; Corrĕa et al. 2017; Wang et al. 2017b; Kovács et al. 2018; Morales et al. 2018). Currently, 431 secondary metabolites, over 380 terpenoids, and 30 steroids (lanostane triterpenes as ganoderic and lucidenic acids, meroterpenoids, pentacyclic triterpenes, prenyl hydroquinone sesquiterpenoids, steroids), as well as alcohols, aldehydes, alkaloids, esters, glycosides, ketones, and lactones with significant bioactivities, were isolated from 22 *Ganoderma* species (*G. amboinense*, *G. annulare*, *G. applanatum*, *G. australe*, *G. boninense*, *G. capense*, *G. carnosum*, *G. cochlear*, *G. colossum*, *G. concinna*, *G. fornicatum*, *G. hainanense*, *G. lipsiense*, *G. mastoporum*, *G. neo-japonicum*, *G. orbiforme*, *G. pfeifferi*, *G. resinaceum*, *G. sinense*, *G. theaecolum*, *G. tropicum*, and *G. tsugae*) (Baby et al. 2015). The structure-activity relationship of 71 triterpenoids and steroids isolated from *Ganoderma* species was revealed (Castellano and Torrens 2015). The chemopreventive agents for cancer, a new lanostanoid tsugaric acid F, and a novel palmitamide with antioxidant and weak cytotoxic activities against PC3 cells were isolated and characterized from the fruiting bodies of *G. tsugae* (Lin et al. 2016). Twelve new highly oxygenated lanostane nor-triterpenoids and 9 known ganoderic acids with moderate inhibitory effects against α-glucosidase were isolated from the fruiting body of *G. lucidum* (Zhao et al. 2015), while 6 new lanostane-type triterpenoids, leucocontextins, and 12 compounds with toxicity

against K562, SMMC-7721, and MCF-7 cells were identified from the fruiting bodies of *Ganoderma leucocontextum* (Zhao et al. 2016). Fourteen lanostane triterpenoids, including nine *Ganoderma* acids and five *Ganoderma* alcohols, were isolated from the fruiting bodies of *Ganoderma hainanense*. Considering that *G. hainanense*, similar to *G. lucidum* and *G. sinense*, contains lanostane triterpenoids, it could also possess a broad spectrum of activities, particularly against HL-60, SMMC-7721, A-549, and MCF-7 cells (Peng et al. 2015). Three new nortriterpenes, ganoboninketals, and highly complex polycyclic systems with antiplasmodial activity against *Plasmodium falciparum*, as well as with weak cytotoxicity against A549 and HeLa cells, and no inhibitory activity against the lipopolysaccharide (LPS)-induced macrophages were detected in other *Ganoderma* species – *G. boninense* (Ma et al. 2014b). Sixteen new lanostane triterpenes (ganoleucoins A-P) together with ten known triterpenes were originally isolated from the cultivated fruiting bodies of *G. leucocontextum* belonging to *G. lucidum* complex (Zhou et al. 2015). These compounds showed inhibitory effects on HMG-CoA reductase and α-glucosidase, as well as cytotoxicity against the K562 and PC-3 cell lines (Wang et al. 2015b). The triterpene lactones (colossolactone G), seven new triterpene lactones (ganodermalactones A–G), five known triterpene lactones (colossolactone B, colossolactone E, colossolactone I, colossolactone IV, and schisanlactone B), and ergosterol have been isolated from mycelial biomass of *Ganoderma* strain KM01 (Lakornwong et al. 2014). Among these biomolecules, several compounds exhibited antimalarial activity against *P. falciparum*.

Fifteen undescribed and five known lanostane-type C31 triterpenoid derivatives (palustrisoic and polyporenic acids) were isolated from the aqueous-ethanolic extract of the fruiting bodies of cultivated *Fomitopsis palustris*. Strong cytotoxicity of polyporenic acid B against HCT116, A549, and HepG2 cell lines and weak cytotoxicity of palustrisolides A, C, and G were revealed (Zhao et al. 2018). Eight undescribed lanostane triterpenoids, pardinols A–H, and one previously reported lanostane triterpenoid saponaceol B were isolated from the fruiting bodies of agaricomycetous species *Tricholoma pardinum* (Zhang et al. 2018). Pardinols B and pardinols E–H exhibited certain inhibitory effects of nitric oxide (NO) production with IC_{50} value ranging from 5.3 to 14.7 mM, as well as cytotoxicity against human cancer cell lines.

It is known that a high rate of aerobic glycolysis which occurs in malignant tumors is one of the most fundamental metabolic alternatives during tumor development and progression (Patra et al. 2013; Ros and Schulze 2013; Tan and Miyamoto 2015). The enzyme hexokinase 2 (HK2) plays a pivotal role in the glycolytic pathway of cancer cells, promotes tumor progression in animal models, and provides a new target for cancer therapy. A new steroid isolated from *Ganoderma sinense* possesses a high binding affinity to HK2 with significant binding free energy. It was identified as an HK2 inhibitor and can be considered as a potential drug targeting the HK2 for cancer therapy (Bao et al. 2018). Highly oxygenated lanostane-type triterpenoids (OLTT) obtained from the fruiting bodies of *Ganoderma gibbosum*, their bioactivity (Pu et al. 2017), and chemotaxonomic significance (Welti et al. 2015) have recently been investigated.

Nine new sesquiterpenoids, clitocybulol derivatives, clitocybulols G–O, and three known sesquiterpenoids, clitocybulols C–E, with moderate inhibitory effects against protein tyrosine phosphatase 1B (PTP1B) were isolated from the solid culture of edible fungus *Pleurotus cystidiosus* (Tao et al. 2016). The piptolinic acid A isolated from a methanolic extract of the fruiting bodies of *P. betulinus* showed cytotoxic activity against human promyelocytic leukemia cell line HL-60 and human acute monocytic leukemia cell line THP-1 (Tohtahon et al. 2017). Fungal bioactive sesquiterpenoid eremophilanes possessing antibacterial, anti-inflammatory, anti-obesity, antiviral, and cytotoxic effects were reported in several mushrooms, particularly belonging to genera *Xylaria* (Yuyama et al. 2017a). Alliacane sesquiterpenoids were isolated from submerged cultures of *Inonotus* sp. (Isaka et al. 2017). The metabolites and multicomponent pharmacokinetics of ergostane and lanostane triterpenoids were identified in the mushroom *Antrodia cinnamomea* (Qiao et al. 2015). Further chemical screening of mushrooms allows discovering new pharmacologically promising terpenoids, steroids, and sterols with therapeutic effects.

1.3.3 Phenolics and Other Compounds

Phenols are a diverse group of biocompounds which include a large number of subclasses, such as flavonoids, phenolic acids, quinones, tocopherols, tannins, etc. Fungal phenolic derivatives are primarily known for their anticarcinogenic, anti-inflammatory, antioxidant, and antimutagenic effects (Palacios et al. 2011; Kozarski et al. 2015; Islam et al. 2016). It has been found that mushroom phenolics are outstanding antioxidants which lack mutagenic properties (Khatua et al. 2013).

The extraction of total phenolics and flavonoids from wild and cultivated edible, medicinal mushrooms by different solvents was reported (Abugri and McElhenney 2013). Phenolic compounds, flavonoids, ascorbic acid, β-carotene, and lycopene were detected in a methanol-soluble extract from fruiting bodies of *G. frondosa* and *Volvariella volvacea* (Acharya et al. 2015, 2016). Five phenolic acids (vanillic acid, *m*-hydroxybenzoic acid, *o*-hydroxybenzene acetic acid, 3-hydroxy-5-methyl benzoic acid, and *p*-hydroxybenzoic acid) with analgesic effects were isolated from *S. commune* (Yao et al. 2016).

The analysis of phenolic constituents, as well as antimicrobial and anti-radical activities of edible mushrooms growing in Poland, have recently been reported (Nowacka et al. 2014). Bioactivities and phenolic contents of 25 wild edible mushrooms from Northeastern Thailand were analyzed for their antioxidant properties, proteins, sugars, β-glucan, and phenolic profiles (Butkhup et al. 2018). The strongest scavenging activity (83.07 and 86.6%) and reductive power (9.79 and 8.42 g Fe^{2+}/kg) were revealed in *Termitomyces clypeatus* and *V. volvacea*, respectively. Both species were identified as sources of healthy compounds (β-glucans and flavonoids) which could be used to mitigate diseases involving free radicals.

The antioxidant capacity and total phenolic contents of wild *T. versicolor* and *T. gibbosa* were evaluated in Romania (Pop et al. 2018). Twenty-eight compounds

were identified as coumarins, flavanols, flavones, flavonols, isoflavonoid derivatives, and phenolic acids. The methanolic extract revealed the highest antioxidant activity, while the highest total polyphenol and flavonoid contents were detected in a water extract of mushrooms. It was suggested that *Trametes* species could be considered a source of bioactive phenolics.

Recent studies of antioxidant activity of cultivated (*A. bisporus*, *P. ostreatus*) and wild-growing (*Agaricus campestris*, *B. edulis*, *C. cibarius*, *Macrolepiota procera*, *P. ostreatus*, *Russula alutacea*, and *R. vesca*) edible mushrooms identified them as sources of bioactive phenolics and flavonoids (Buruleanu et al. 2018).

Fungal alkaloids (N-containing heterocyclic compounds) are known for their toxicological relevance, e.g., ergot alkaloids in *Claviceps purpurea* and psilocybin in *Psilocybe* species (Liu 2005; Streith 2011; Tylš et al. 2014). Eight new alkaloids isolated from the medicinal mushroom *H. erinaceus* with inhibitory effects against protein tyrosine phosphatase-1B (PTP1B), α-glucosidase and moderate cytotoxicity against K562 cells were described (Wang et al. 2015a).

The nucleic acid constituents (purine and pyrimidine nucleobases, uridine, guanosine, adenosine, and cytidine nucleosides) play an important role in the regulation of various physiological processes in the human body. The nucleic acid constituents and other bioactive compounds (alkaloids, polysaccharides, and terpenoids) were recently identified in several edible and nonedible mushrooms (Phan et al. 2017b, 2018).

1.4 Pharmacological Activity of Mushrooms

The scientific research and case studies from a traditional medicine show that mushrooms possess promising pharmacological potential. Bioactive compounds and extracts from medicinal mushrooms showed mainly anti-allergic, antibacterial, antidepressant, antifungal, anti-inflammatory, antioxidant, antiviral, cardioprotective, hepatoprotective, neuroprotective, cytotoxic, hypotensive, and immunomodulatory activities (Badalyan 2003a, b, 2004a, b, 2012; Lindequist et al. 2005; Fan et al. 2006; Wasser 2010, 2014; Chang and Wasser 2012; Roupas et al. 2012; Stachowiak and Reguła 2012; De Silva et al. 2013; Ivanova et al. 2014; Paterson and Lima 2014; Muszynska et al. 2015; Xu et al. 2016a, b; Sharma et al. 2017; Özcan and Ertan 2018). Therefore, mushrooms can be considered as prospective sources of new myco-pharmaceuticals or natural product-derived drugs.

Several mushrooms (*A. brasiliensis*, *Agrocybe cylindracea*, *A. auricula-judae*, *Coprinus comatus*, *G. applanatum*, *G. lucidum*, *G. frondosa*, *Gymnopus confluens*, *H. erinaceus*, *Inonotus obliquus*, *Ophiocordyceps* (syn. *Cordyceps*) *sinensis*, and *Tremella fuciformis*) are used in the production of functional foods for the prevention and treatment of diabetes mellitus (Perera and Li 2011). Triterpenes isolated from the medicinal mushroom *Wolfiporia cocos* also reduce blood glucose levels in mice (Sato et al. 2002).

One of the most valuable medicinal fungi in China is ascomycetous mushroom *O. sinensis* known for its invigorating effects in strengthening the body and

restoring energy. The fungus parasitizes larvae of moths and converts them into sclerotia from which grows the fruiting body of the fungus (Fig. 1.2b). A considerable effort has been devoted to the study of host insects related to the fungus. The information on the management of insect resources and sustainable use of *O. sinensis* has been reported (Wang and Yao 2011). The extract derived from mycelia and ascomata of *Cordyceps sinensis* and *C. militaris* possess anticancer effects by modulating the immune system and inducing cell apoptosis (Khan et al. 2010; Xu et al. 2016b). The availability of laboratory cultures of these entomopathogenic fungi will further expand their pharmacological usage (Wang et al. 2012).

Another well-known medicinal mushroom *A. subrufescens* contains a number of bioactive compounds with immunomodulatory, anticancer, and other pharmacological effects (Ahn et al. 2004; Kim et al. 2005; Ellertsen and Hetland 2009; Johnson et al. 2009; Bouike et al. 2011; Lima et al. 2011; Wang et al. 2013; Da Silva de Souza et al. 2017) (Fig. 1.1a). We recommend following Kerrigan (2005, 2016), Ludwig (2007), Cappelli (2011), Parra (2013), and Thongklang et al. (2014, 2017) for taxonomy and synonymy of *A. subrufescens*. The main synonyms of *A. subrufescens* are *Agaricus blazei* Murrill sensu Heinemann, *A. rufotegulis* Nauta, *A. brasiliensis* Wasser, M. Didukh, Amazonas and Stamets, *A. albopersistens* Zuccher, and *A. bambusae* Beeli var. *bambusae*. Previously, *A. subrufescens* has not only been misidentified as *A. blazei* but also to *A. sylvaticus* Schaeff. (Souza Dias et al. 2004). In other respects, Kerrigan (2005) hybridized between each other strains isolated from North, South America, Brazil, France, and Thailand (Thongklang et al. 2014, 2016).

The compounds extracted from edible medicinal *Pleurotus* mushrooms showed efficacy for treatment of several chronic disease (Gunde-Cimmerman 1999) due to their antibacterial, anti-inflammatory, anti-mitogenic, antitumor, antioxidant, antiviral, hypoglycemic, immunomodulatory, cardioprotective, and other pharmacological activities (Jose et al. 2002; Filipic et al. 2002; Hossain et al. 2003; Hu et al. 2006; Badalyan et al. 2008b; Baggio et al. 2010; Papaspyridi et al. 2011; Patel et al. 2012; Schillaci et al. 2013; Jayasuriya et al. 2015; Khatun et al. 2015; Fu et al. 2016; Venturella et al. 2016; Zhang et al. 2016; Acharya et al. 2017; Adebayo et al. 2018; Baskaran et al. 2017; Debnath et al. 2017; Ebrahimi et al. 2017; Masri et al. 2017; Owaid et al. 2017; Abidin et al. 2018; Finimundy et al. 2018a, b).

Bioactive compounds extracted from inedible bracket fungi also possess pharmacological potential and could be used to develop healthcare products with different formulations (Barros et al. 2008; Dai et al. 2009; Reis et al. 2014; Heleno et al. 2015a, b). The bioactive compounds isolated from *T. versicolor* exhibited health-beneficiary effects, as inhibitors of aflatoxins (Scarpari et al. 2016), while compounds (amino acids, aromatic acids, flavones, polysaccharides, triterpenes, etc.) isolated from *Ph. linteus* showed anticancer, hypoglycemic, anti-inflammatory, antioxidant, and immunoregulatory activities (Chen et al. 2016). The induction of ovulation by the extract of edible Maitake mushroom (*G. frondosa*) in patients with polycystic ovary syndrome (Chen et al. 2010) and medicinal properties of *F. fomentarius* and *F. pinicola* have been reported (Badalyan and Shahbazyan 2015).

A comparative chemical analysis, acute toxicity, and pharmacological activity of fruiting body extracts of several agaricoid mushrooms, particularly of nonedible

Nematoloma species, poisonous species *Cortinarius armillatus* and *T. pardinum*, as well as of edible species *Flammulina velutipes* and *Paxillus involutus*, were performed (Badalyan et al. 1994, 1995, 1996, 1997a, b, 2001). The antioxidant activity (Badalyan 2003b), cytokine induction, and immunomodulatory effect of fruiting body extract of Enokitake mushroom *F. velutipes* (Badalyan and Hambardzumyan 2001), antiprotozoal activity, and mitogenic/regenerative effects of mycelia of shiitake mushroom *L. edodes* (Badalyan 2004b) and other culinary-medicinal *Agaricomycetes* species (Badalyan 2003a) have also been reported. The latest advancements in the study of the pharmacological potential of mushrooms are further discussed in this chapter.

1.4.1 Antimicrobial (Antifungal and Antibacterial) Activity

Agaricoid and polypore mushrooms are known as active producers of antimicrobial (antibacterial and antifungal) compounds. Many of these compounds, such as velutin and flammulin from *F. velutipes*; applanoxidic acid, ganodermadiol, ganomycin, and ganoderiol from *G. lucidum*; lentinan from *L. edodes*; schizophyllan from *S. commune*; and others, have been shown to possess antifungal and antibacterial properties (Wasser and Weis 1999; Suay et al. 2000; Badalyan 2004a, b, 2008a; Badalyan et al. 2002, 2004; Lindequist et al. 2005; Dyakova et al. 2011; Alves et al. 2012; Khatun et al. 2012; Mohanarji et al. 2012; Nowacka et al. 2014; Ferreira et al. 2015; Heleno et al. 2015b; Jaszek et al. 2015; Bedlovičová et al. 2016; Canli et al. 2016; Ćilerdžić et al. 2016; Kosanić et al. 2016; Falade et al. 2017; Gargano et al. 2017; Kolundčzić et al. 2017; Morris et al. 2017; Shen et al. 2017; Waithaka et al. 2017; Chepkirui et al. 2018; Özcan and Ertan 2018).

In vitro screening of methanolic extracts from fruiting bodies of 28 Yemeni basidiomycetous mushrooms revealed antibacterial and antifungal effects against Gram-positive (*Bacillus subtilis*, *Micrococcus flavus*, *Staphylococcus aureus*) and Gram-negative (*Escherichia coli*, *Pseudomonas aeruginosa*) bacteria, as well as fungal pathogens, including dermatophytes, molds, and yeasts (*Aspergillus fumigatus*, *Candida albicans*, *C. krusei*, *Microsporum gypseum*, *Mucor* sp., and *Trichophyton mentagrophytes*) (Al-Fatimi et al. 2013). The highest antibacterial activity was observed in *Agaricus bernardii*, *Agrocybe pediades*, *Chlorophyllum molybdites*, *Coriolopsis polyzona*, *Ganoderma xylonoides*, *Pycnoporus sanguineus*, *Trametes cingulate*, and *T. lactinea* species, while the extracts from *C. molybdites*, *G. xylonoides*, and *P. sanguineus* possessed significant antifungal activity.

Antifungal and antibacterial activities against *B. subtilis*, *Ceratocistys pilifera*, *E. coli*, *E. faecalis*, *Fusarium oxysporum*, *Penicillium notatum*, *P. aeruginosa*, *Rhizoctonia solani*, and *S. aureus* were evaluated in 26 compounds isolated from cultural broth of *Collybia butyracea*, *Entoloma nubigenum*, *Inocybe geophylla*, *Mycena hialinotricha*, *Psathyrella* species, and *Stropharia semiglobata* mushrooms collected from sub-Antarctic forests in Southern Chile (Reinoso et al. 2013). Based on the results, tested mushrooms have been recommended for further studies as sources of antimicrobial compounds.

Biocontrol properties and the potential role of basidiomycetous mushrooms as antiphytobacterial, antiphytofungal, antiphytoviral, larvicidal, and nematicidal agents were recently described by Sivanandhan et al. (2017). The screening of antiphytopathogenic effects in 20 polypore mushrooms against phytopathogens (e.g., *Bipolaris sorokiniana*, *Fusarium culmorum*, *F. oxysporum*, *Gaeumannomyces graminis* var. *tritici* (syn. *Gaeumannomyces tritici*), *Ophiostoma ulmi*, *Pestalotiopsis funerea*, *Rhizoctonia cerealis*) and their antagonists (*Gliocladium roseum*, *Trichoderma asperellum*, *T. harzianum*, *T. pseudokoningii*, and *T. viride*) revealed the highest activity in white rot species *Climacodon septentrionale*, *F. fomentarius*, *Ganoderma resinaceum*, and *Lentinus tigrinus* (Badalyan and Gharibyan 2017b).

The methanolic and aqueous extracts from sclerotia of the wild-growing mushroom *L. rhinocerotis* were tested against 15 pathogenic bacteria from genera *Bacillus*, *Corynebacterium*, *Escherichia*, *Klebsiella*, *Micrococcus*, *Pseudomonas*, *Salmonella*, *Staphylococcus*, and *Streptococcus* and 4 fungi from genera *Candida* and *Mucor* (Mohanarji et al. 2012). The extracts (30 mg/mL) showed a significant inhibition against all test organisms, except for *Streptococcus pyogenes* and *Serratia marcescens*. The chemical analysis of extracts revealed the presence of alkaloids, proteins, gums, mucilage and flavonoids.

The antimicrobial activity of extracts obtained from fruiting bodies of *A. bisporus* and *T. gibbosa*, collected in Kenya, against several Gram-positive and Gram-negative bacteria, as well as phytopathogenic and keratinophilic fungi, was reported (Waithaka et al. 2017). The concentration-dependent bacteriostatic and bacteriocidic effects against *Helicobacter pylori* strain were revealed in ethanolic extracts from *A. bisporus*, *A. brasiliensis*, *Agrocybe aegerita*, *C. comatus*, *C. militaris*, *F. velutipes*, *G. lucidum*, *G. frondosa*, *H. erinaceus*, *Hypsizygus marmoreus*, *L. edodes*, *Ph. igniarius*, *P. eryngii*, and *P. ostreatus* (Shang et al. 2013). However, further research is needed to identify the role of active compounds of tested mushrooms in the treatment of *H. pylori*-associated gastrointestinal disorders.

Previously seven terpenoids named microporenic acids (A-G) isolated from the cultures of polypore *Microporus* species showed antibacterial activity against biofilm formation by Gram-positive *S. aureus* bacteria, as well as antifungal activity against *Candida tenuis* and *C. albicans* (Chepkirui et al. 2018). The extracts obtained from wild mushrooms *Fistulina hepatica*, *Lepista nuda*, *Leucopaxillus giganteus*, *Mycena rosea*, and *Russula delica* were originally reported as antimicrobials against in vitro biofilm formation by multiresistant clinical isolates of Gram-positive bacteria (*Acinetobacter baumannii*, *E. coli*, *Proteus mirabilis*, and *P. aeruginosa*) to solve multidrug resistance problems in public healthcare (Alves et al. 2014). The antibacterial and anti-biofilm compounds were originally isolated from *Hypoxylon fragiforme* mushroom (Yuyama et al. 2017b).

In vitro antibacterial, antifungal, and antioxidant properties of mycelial and fruiting body extracts obtained from several agaricomycetous fungi, such as *G. applanatum* (Klaus et al. 2016), *Pleurotus* spp. (Owaid et al. 2017), *Pycnoporus sanguineus* (Jaszek et al. 2015), *Rigidoporus microporus* (Falade et al. 2017), *Xylaria hypoxylon* (Canli et al. 2016), and other species, were recently reported (Dyakova et al. 2011; Ren et al. 2014; Ferreira-Silva et al. 2017; Gaylan et al. 2018).

The aqueous extracts of edible desert truffles *Picoa juniperi*, *Terfezia claveryi*, and *Tirmania pinoyi* (phylum *Ascomycota*) showed in vitro antibacterial activity against the Gram-positive human pathogenic reference strain *S. aureus* ATCC 29213 and the Gram-negative *P. aeruginosa* strain ATCC 15442. The acid-soluble protein extracts of *T. pinoyi* and *T. claveryi* showed minimum inhibitory concentrations (MIC) of 50 μg/mL against the tested pathogens (Schillaci et al. 2017).

Further studies directed to the mechanisms behind antimicrobial activities of mushroom-derived compounds are needed.

1.4.2 Immunomodulatory and Anticancer Activities

The immunomodulatory activity is the most prominent pharmacological property of medicinal mushrooms (Badalian 2000; Borchers et al. 2008; Novak and Vetvicka 2008; Ferreira et al. 2010; Tangen et al. 2015; Ahmad et al. 2016; Del Buono et al. 2016; Xu et al. 2016a; Diling et al. 2017; Wasser 2017; Rubel et al. 2018; Wang et al. 2018).

Mushroom derived bioactive molecules, particularly β-glucans schizophyllan, lentinan, and grifolan, proteins, glycoproteins, and lipopolysaccharides (LPS), have been identified as immunomodulators and are widely used in the treatment of several types of cancer (Keong et al. 2007; Wasser 2017). They can prevent oncogenesis by possessing direct inhibitory effects on tumor metastasis and exhibit antitumor effects by inducing immune response in the host (Wasser and Weis 1999; Wasser 2002, 2017; Brown and Gordon 2003; Wasser and Didukh 2005; Oba et al. 2007; Novak and Vetvicka 2008; Chan et al. 2009; Volman et al. 2010; Adotey et al. 2011; Aleem 2013; Mizuno and Nishitani 2013; Yoon et al. 2013; Guggenheim et al. 2014; Ivanova et al. 2014; Khan et al. 2014; Kozarski et al. 2014; Ma et al. 2014a; Dai et al. 2015; Fritz et al. 2015; Del Buono et al. 2016; Tsai et al. 2016; Gargano et al. 2017; Glamočlija and Soković 2017; Hapuarachchi et al. 2017; He et al. 2017a; Wang et al. 2017a; Pandya et al. 2018).

Mushroom-derived glucans belong to PAMPs (pathogen-associated molecular patterns) and after per os administration are recognized by PRR (pattern recognition receptors) on the surface of dendritic cells and macrophages in the gastrointestinal tract (Muta 2006; Batbayar et al. 2012; Wasser 2017; Patin et al. 2018). After recognition by PRR, the glucan molecules are internalized into and fragmented within the cells. The fragments are taken up by the lymph and transported to other parts of the immune system. They bind to specific CR3 receptors of the complement system on the surface of immune cells (neutrophil granulocytes and NK cells) and activate them which is followed by secretion of cytokines, such as tumor necrosis factors alfa (TNF-α), interferons (IFN-γ), and interleukins (IL-6, IL-8, IL-12), and leads to activation of cytotoxic T lymphocytes, T helper cells, and B cells. Increased phagocytosis, production of NO, formation of antibodies occur, and innate and adaptive immunity are activated which react against invading microorganisms and abnormal cells, such as tumor cells (Wasser 2002, 2017; Brown and Gordon 2003; Chen and Seviour 2007; Novak and Vetvicka 2008; Chan et al. 2009; Barsanti et al. 2011; Batbayar et al. 2012; Ren et al. 2012; Giavasis 2014; Guggenheim et al. 2014).

It is known that immunomodulatory effects depend on biological conditions of the organism. The immunomodulatory activity of pharmaceuticals also depends on the dosage, mode, timing of administration, and on pharmaceutical formulation (Wasser 2002, 2017).

The immune stimulation by fungal β-glucans can be used in the adjunct therapy of cancer patients together with surgery, chemotherapy, and radiation treatment (Oba et al. 2007; Ramberg et al. 2010; Zong et al. 2012; Guggenheim et al. 2014; Twardowski et al. 2015) (see part 6).

Apart from immunomodulation, β-glucans possess also direct tumoricidal effects by inhibiting the expression of aromatase, an enzyme responsible for the conversion of androgens to estrogens which is often upregulated in breast cancer patients (Adams et al. 2008). Furthermore, fungal β-glucans may modify cell cycle-regulating genes, arrest the cell cycle, and induce apoptosis (Jiang and Sliva 2010). Maitake's (*G. frondosa*) β-glucan stimulates differentiation of haematopoietic progenitor cells, production of granulocyte colony-stimulating factor, and the recovery of peripheral blood leukocytes after bone marrow injury in phase II study involving patients with a preleukemic myelodysplastic syndrome (Wesa et al. 2015). These compounds change adhesion molecules on the surface of cancer cells; inhibit migration, invasion, and adhesion of cells; and possibly affect the development of metastases (Masuda et al. 2008; Jiang and Sliva 2010). A recent study investigated the effect of Maitake D-fraction to attenuate the aggressive activity of triple-negative breast cancer (TNBC) cells which revealed that the proteoglucan fraction induces MDA-MB-231 cell apoptosis, reduces their metastatic potential, and decreases invasive capacity (Alonso et al. 2018).

The polysaccharide fucogalactan isolated from the aqueous extract of giant mushroom *Macrocybe titans* showed anticancer activity against murine melanoma cells B16-F10 by reducing their migration in vitro, however did not alter the viability, proliferative capacity, and morphology of cells (Da Silva Milhorini et al. 2018). The 14,942 Da polysaccharide, isolated from *Collybia radicata* with potent immunomodulatory activity, may be considered a novel immunomodulator to be used in medical and food industries (Wang et al. 2018).

The therapeutic effects of β-glucans against oncological diseases are supported by low-molecular-weight compounds of mushrooms, such as triterpenes, isolated from *G. lucidum*, and other *Ganoderma* species, *A. brasiliensis*, *I. obliquus*, and *W. cocos*. These compounds exhibit immunomodulatory, cytotoxic, and apoptosis-inducing effects by inhibiting cell cycle and phosphorylation of Erk1/2, increasing the level of p53 and *Bax*, upregulating NF-κB and AP-1, and reducing the activity of topoisomerase II (Ríos et al. 2012; Rubel et al. 2018).

Methanolic extracts of 29 different wild edible mushrooms were examined for antioxidant, antiproliferative, cytotoxic, and pro-apoptotic activities toward a lung adenocarcinoma cell line A549 (Vasdekis et al. 2018). Certain species exhibited a high antioxidant activity which was related to their high total phenolic content, whereas *C. cibarius*, *Cantharellus cinereus*, *Craterellus cornucopioides*, and *Hydnum repandum* (order *Cantharellales*) showed high cytotoxicity and induced apoptosis in A549 cells. As an active ingredient, piceatannol with antiproliferative

activity was originally identified in studied mushrooms. A novel antitumor glucan extracted from the fruiting bodies of *Coriolus* (syn. *Trametes*) *versicolor* was purified and characterized (Awadasseid et al. 2017). The effects of *Inonotus hispidus* on human immunocompetent cells (Gründemann et al. 2016), anticancer and anti-inflammatory effects of *G. lucidum* on melanoma cells (Barbieri et al. 2017), antiproliferative effects of extract obtained from *P. ostreatus* on acute leukemia cell lines (Ebrahimi et al. 2017), RNA fraction from *B. edulis* against myelogenous leukemia cells (Lemieszek et al. (2017), as well as novel acid polysaccharides (Zhang et al. 2017) and anticancer recombinant latcripin 11 from *L. edodes* (Gao et al. 2018) were recently reported. A ganoderic acid with antiproliferative activities from neotropical *Ganoderma* mushrooms (Welti et al. 2010) and a recombinant fungal immunomodulatory protein (FIP) from *Ganoderma atrum* inducing apoptosis in breast cancer cells (Xu et al. 2016a) were characterized.

Seven steroids based on ergostane, as well as lanostane and ceramide derivatives, were originally isolated and identified from the methanolic extract of agaricomycetous mushroom *Scleroderma bovista* (Kovács et al. 2018). The lanostane derivatives and ergosterol peroxide 3-glucoside revealed significant antiproliferative properties on one or more tested human cancer cell lines (HeLa, A2780, MDA-MB-231, and MCF-7).

The antitumor effects of a new pyrrole compound By-1 obtained from the submerged culture of medicinal mushroom *Taiwanofungus camphoratus* have recently been revealed against various cancer cells (Yang et al. 2018). It has been shown that combinations of pro-apoptotic and anti-autophagy agents from *T. camphoratus* could be an effective new treatment strategy for non-small cell lung adenocarcinoma.

Further studies of the signaling pathways modulating cancer development and progression will promote the transfer of bioactive mushroom molecules to clinically effective therapeutic agents (Aras et al. 2018).

1.4.3 Hypocholesterolemic, Hypoglycemic, and Anti-obesity Effects

Hyperglycemia, hyperlipidemia, insulin resistance, obesity, and arterial hypertension are important symptoms of metabolic syndrome, preceding signs of type 2 diabetes and risk factors of cardiovascular diseases (CVD) (Francia et al. 1999; Gunde-Cimmerman 1999; Lo and Wasser 2011; De Silva et al. 2012b; Martel et al. 2017; Vitak et al. 2017; Morales et al. 2018). The existing drugs, such as insulin, statins, and angiotensin-converting enzyme (ACE) inhibitors used for the treatment of metabolic syndrome, have limited therapeutic efficacy and several side effects. There are other medications (e.g., HMG-CoA reductase, aldose reductase, and α-glucosidase) which are also used for the treatment of metabolic syndrome. Nevertheless, considerable effort has been made in the pharmaceutical industry to develop new preparations to improve glucose and lipid metabolism without significant side effects.

Mushrooms and mushroom-rich nutrition are regarded as healthy dietary food supplements to prevent and treat such widespread pathological conditions (Guillamón et al. 2010). The results of in vitro animal assays and several human studies suggest that medicinal mushrooms *A. bisporus* (Yamac et al. 2010), *A. brasiliensis* (Kim et al. 2005), *G. lucidum* (Ma et al. 2015), *G. frondosa* (Hong et al. 2007), *H. erinaceus* (Thongbai et al. 2015), *Ph. linteus* (Yamac et al. 2016), and *Pleurotus* species (Jayasuriya et al. 2015; Anjana and Savita 2017) can normalize blood glucose and lipid levels. Several edible and nonedible mushrooms, such as *L. edodes* (Poucheret et al. 2006; De Silva et al. 2013; Yang et al. 2013) and *Hypholoma fasciculare* (Badalian and Serrano 1999), have revealed beneficial effects on lowering the levels of blood glucose, lipids, and arterial pressure in experimental animals. Moreover, the eritadenine extracted from *L. edodes* has been identified as an anti-atherogenic compound which not only improves lipid metabolism but also inhibits the activity of ACE in vitro (Afrin et al. 2016).

Several new bioactive compounds, including lanostane triterpenoids (e.g., lucidenic acids, ganolucinins, new natural product ganomycin, etc.) isolated from fruiting bodies of *G. lucidum*, were suggested as promising bioactive agents for the treatment of metabolic syndrome (Chen et al. 2017). The obtained results provide evidence for the usage of *G. lucidum* extract as a myco-medicine and food supplement to control hyperglycemia and hyperlipidemia.

Evaluation of antidiabetic properties of agaricomycetous culinary-medicinal mushrooms *Calocybe indica*, *P. ostreatus*, and *V. volvacea* in mice revealed that *C. indica* could be identified as a natural source of antidiabetic compounds (Singh et al. 2017). The aqueous extract of bracket fungi *Inocutis levis* improves insulin resistance and glucose tolerance in high sucrose-fed Wistar rats (Ehsanifard et al. 2017). The hypoglycemic effect of *Pleurotus giganteus* is manifested by enhancing adipocyte differentiation and glucose uptake via activation of peroxisome proliferator-activated receptor (PPAR) and glucose transporters 1 and 4 in 3T3-L1 cells (Paravamsivam et al. 2016).

In vitro anti-atherogenic effects of extracts derived from *P. pulmonarius* based on inhibition of ACE and HMG-CoA reductase and the protective effects of extracts on the endothelial membrane against oxidative stress were recently evaluated (Abidin et al. 2018). The results of this study could be applied in nutriceutical and pharmaceutical industries.

Literature data and advancements on the treatment of type 1 diabetes mellitus using medicinal mushrooms were analyzed by Vitak et al. (2017). However, data concerning molecular mechanisms of action of hypocholesterolemic and hypoglycemic compounds obtained from edible mushrooms are still rare (Gil-Ramirez et al. 2017).

1.4.4 Antioxidant and Anti-inflammatory Effects

The maintenance of balance between free radical production and antioxidant defense is an essential condition for normal functioning of biological organisms.

Free radicals may damage cellular lipids, proteins, and DNA, affecting their normal function and leading to various diseases. Natural products with antioxidant activity may help the endogenous defense system, and from these perspective antioxidants present in diet are considered possible protective agents reducing oxidative damage. In this regard, mushrooms, including edible and medicinal species, are regarded as a natural source of antioxidants (Saltarelli et al. 2009, 2015; Palacios et al. 2011; Chang and Wasser 2012; Khatua et al. 2013; Ferreira et al. 2015; Jaworska et al. 2014; Heleno et al. 2015a; Khatun et al. 2015; Ćilerdžić et al. 2016; Bandara et al. 2017; Debnath et al. 2017; Souilem et al. 2017; Carocho et al. 2018; Choi et al. 2018; Da Silva de Souza et al. 2018; Khaskheli et al. 2018; Morel et al. 2018; Anwar et al. 2018). Mushrooms are also described as natural sources of anti-inflammatory agents (Jose et al. 2002; Ruthes et al. 2013; Lee et al. 2014a; Taofiq et al. 2016a, c; Souilem et al. 2017).

The antioxidant and anti-inflammatory compounds identified in mushrooms include fatty acids, phenolic acids, polysaccharides, steroids, terpenes, tocopherols, and other biomolecules. Among them, phenolics, polysaccharides, and terpenoids are considered the determinants of antioxidant and anti-inflammatory effects. However, further clinical studies are needed out to confirm the antioxidant and anti-inflammatory potential of these compounds (Taofiq et al. 2016a).

The antioxidant properties of wild mushrooms have been extensively investigated by Ferreira et al. (2009). Many antioxidant compounds, such as phenolics, tocopherols, ascorbic acid, and carotenoids, have been identified, and the mechanisms of action of antioxidant properties were discussed. It was suggested that wild or cultivated edible mushrooms could be used as dietary healthy food products, taking advantage of synergistic effects of all present bioactive compounds (polyphenols, polysaccharides, vitamins, carotenoids, and minerals) (Kozarski et al. 2014, 2015).

The study on fermentation broth of three *Ganoderma* species exhibiting a large number of phenol derivatives and flavonoids showed that antioxidant activity of *G. lucidum* was higher than the activity of *G. applanatum* and *G. carnosum* (Ćilerdžić et al. 2016). Moreover, *G. lucidum* strains were the most effective antibacterial and antifungal agents. The cultural broth of tested *Ganoderma* species was suggested as a potent natural antioxidant and antimicrobial agent.

Antioxidant potential of an aqueous extract from *A. brasiliensis* was evaluated in adjuvant-induced arthritic rats (Da Silva de Souza et al. 2018). The antioxidant defenses, diminished by arthritis, were improved by treatment with *A. brasiliensis* extract. Aqueous preparations of *A. brasiliensis* can be recommended as potential auxiliaries in the treatment of patients with rheumatoid arthritis due to their capacity of reducing oxidative stress.

A potent antioxidant effect was detected in methanolic extracts from mycelia of ten *Pleurotus* species, including *P. citrinopileatus* and *P. sajor-caju.* The inclusion of these mushrooms in diet may help to prevent diseases caused by oxidative damage (Debnath et al. 2017). Screening of standardized hydroalcoholic extracts of four *Pleurotus* species (*P. levis*, *P. ostreatus*, *P. pulmonarius*, and *P. tuber-regium*) revealed more remarkable antioxidant and antibacterial properties in *P. levis* and *P.*

tuber-regium compared to other studied species that are commercially cultivated (Adebayo et al. 2018). It was shown that *Pleurotus* mushrooms *P. citrinopileatus*, *P. florida*, and *P. pulmonarius* are comparable to best antioxidants, which is attributed to their catalase, phenolics, and peroxidase contents (Khatun et al. 2015).

Antioxidant and antiproliferative effects of cyclohexane, chloroform, ethanol, and water extracts of 28 wild *Boletales* mushrooms (*Caloboletus calopus*, *Gyroporus castaneus*, *Omphalotus olearius*, *Rubroboletus lupinus*, *R. pulchrotinctus*, *R. satanas*, and *Suillus luteus*) from France were recently evaluated (Morel et al. 2018). Among the tested mushroom extracts, 11 presented antiproliferative activities against HCT116 cells, while the highest antioxidant effects were revealed for chloroform, ethanol, or aqueous extracts, depending on mushroom species. Further studies are necessary to identify bioactive compounds and valorize mushrooms either for edible species, as healthy foods, or for inedible mushrooms, as natural ingredients in the pharmaceutical and food industries (Morel et al. 2018).

A strong antioxidant effect was described in methanolic extracts derived from *C. molybdites*, *G. xylonoides*, *Hexagonia velutina*, *P. sanguineus*, *Trametes lactinea*, and *T. cingulate* mushrooms (Al-Fatimi et al. 2013). The extract of the medicinal mushroom *A. auricula-judae* was reported as an antioxidant promoting procollagen biosynthesis in HaCaT cells (Choi et al. 2018). The antioxidant potential and antiproliferative activity were evaluated in edible and medicinal agaricoid (*A. subrufescens*, *B. edulis*, *F. velutipes*, *Ganoderma capense*, *H. erinaceus*, and *Pleurotus djamor*) (Badalyan 2003a, b; Llarena-Hernández et al. 2015; Panthong et al. 2016; Peng et al. 2016; Acharya et al. 2017; Novakovič et al. 2017; Sánchez 2017b; Carocho et al. 2018), as well as in culinary-prepared *S. luteus* (Jaworska et al. 2014) mushrooms.

Fungal polysaccharides play a vital role as dietary free radical scavengers in the prevention of oxidative damage in living organisms (De Silva et al. 2013). The study of chemical characteristics and antioxidant properties of water-soluble and crude polysaccharides obtained from *A. auricula*, *L. edodes*, and *Poria* (syn. *Wolfiporia*) *cocos* revealed that they are composed of β-glycoside linkages (Ke and Chen 2016; Khaskheli et al. 2018). The main compositions of monosaccharides were D-galactose, D-glucose, L-rhamnose, arabinose, and D-mannose for *A. auriculara* and *L. edodes*, while *P. cocos* consisted of D-mannose and D-galactose. The exopolysaccharide (EPS) isolated from agaricomycetous *R. microporus* as a natural antioxidant is recommended for application in food and medicines (Jia et al. 2018).

Antinociceptive and anti-inflammatory effects, as well as structure/activity correlation of (1–3)- and (1–6)-β-D-glucans from *Lactarius rufus* (Ruthes et al. 2013) and induction of pro-inflammatory cytokine production by glucan lentinan from *L. edodes* (Ahn et al. 2017), were reported. The anti-inflammatory potential and molecular mechanisms underlying the protective effect of ethanolic extract of culinary-medicinal mushroom *P. giganteus* against LPS and combination of LPS and hydrogen peroxide-induced inflammation on RAW 264.7 macrophages were revealed (Baskaran et al. 2017).

Data concerning the anti-inflammatory effects of aqueous, ethanolic, and methanolic extracts obtained from fruiting bodies and sclerotia of tiger milk mushroom *L.*

rhinocerotis was recently published (Lee et al. 2014a; Baskaran 2015; Nallathamby et al. 2016, 2018). The ethanolic extract revealed a significant decrease of NO production, unlike the aqueous extract. Moreover, the ethanolic extract was able to stimulate signal transducer and activator of transcription 3 (STAT3) pathways by reducing expression of inducible NO synthase (iNOS) and cyclooxygenase-2 (COX-2) and increasing the expression of interleukin 10 (Baskaran 2015; Nallathamby et al. 2018). The ethyl acetate fraction of the ethanolic extract of *L. rhinocerotis* significantly reduced the NO production in microglial BV2 cells (Nallathamby et al. 2016). The major compounds of ethyl acetate fraction were linoleic and oleic fatty acids and ethyl linoleate. The treatment with linoleic acid significantly decreased the expression of iNOS and COX-2 by 1.2-fold compared to control. Studies of LPS-induced brain microglial BV2 cells pre-treated with hot aqueous extract (500 μg/mL), n-butanol fraction of hot aqueous extract (250 μg/mL), and ethyl acetate fraction of hot aqueous extract (250 μg/mL) of *L. rhinocerotis* showed maximal inhibition of NO production by 88.95, 86.50, and 85.93%, respectively (Seow et al. 2017).

Further studies are needed to elucidate the antioxidant and anti-inflammatory potential of wild and cultivated mushroom resources for their further usage as healthy food supplements and myco-pharmaceuticals.

1.4.5 Cardioprotective Effect

Cardiovascular diseases (CVD) (e.g., heart attack or stroke) affect the heart and circulatory system and are considered the leading cause of death worldwide. The main risk factors for CVD include elevated arterial pressure, high levels of blood glucose and cholesterol. Hypocholesterolemic and hypoglycemic properties of different medicinal mushrooms allow using them as natural, healthy food to prevent the development of diseases and improve cardiovascular health (see parts 4.3, 5, and 6).

Mushrooms, particularly *G. frondosa*, *L. edodes*, and *P. ostreatus*, are almost ideal for low-calorie diets to prevent CVD due to a high content of fiber, proteins, and microelements (Khatun et al. 2007, 2012). Recent experimental and clinical data has demonstrated that many bioactive compounds, i.e., terpenoids, peptides, isoflavones such as biochanin A and formononetin, lanosterone derivative as fomiroid A, and lovastatin extracted from *Boletus aestivalis*, *Clitocybe nuda*, *G. lucidum*, *G. frondosa*, *H. marmoreus*, *L. edodes*, and *Pleurotus* species, can regulate the levels of low, high-density lipoproteins and homocysteine, total cholesterol, and fasting triglycerides and prevent the development of arterial hypertension, oxidative stress, diabetes, and other CVD (Shibu et al. 2017).

Thrombosis can also cause CVD due to fibrin aggregation in the blood (Previtali et al. 2011). Fibrinolysis is the process of dissolving the fibrin in blood clots by proteolytic enzymes. Fibrin clot formation and fibrinolysis are balanced processes in biological systems. However, accumulation of fibrin clots in blood vessels causes thrombosis and leads to myocardial infarction or other CVD.

The tissue-type, the urokinase-type, and the bacterial plasminogen activators, nattokinase and streptokinase, are well-known fibrinolytic agents with a wide range of clinical applications. However, uncontrolled use of these drugs results in side effects (internal hemorrhage, allergic reactions, limitation in specificity toward fibrin, etc.) and is costly (Blann et al. 2002). Therefore, the search for natural sources of fibrinolytic agents is topical.

Mushrooms belonging to different taxonomic and ecological groups are considered active producers of proteolytic (fibrinolytic, thrombolytic, caseinolytic, etc.) enzymes (Denisova 1982, 2010; Denisova et al. 1989; Badalyan et al. 2008b, 2016; Sharjahan et al. 2017). The proteases with fibrinolytic activity were originally isolated from *F. velutipes* (Morozova et al. 1982). Later, other basidiomycetous mushrooms were screened for fibrinolytic and thrombolytic effects (Denisova 1982, 2010; Denisova et al. 1989; Kudryavtseva et al. 2008; Lu and Chen 2012; Badalyan et al. 2016). A number of systematic studies have discovered fibrinolytic proteases from fruiting bodies and mycelia of medicinal mushrooms, such as *Armillariella mellea* (Kim and Kim 1999), *Auricularia polytricha* (Ali et al. 2014, 2017), coprini species (Badalyan et al. 2008b), *C. militaris* (Kim et al. 2006), *F. pinicola* (Badalyan et al. 2016), *Ganoderma* and *Fomes* species (Choi and Sa 2000; Kumaran et al. 2011; Hadda et al. 2015a, b), *H. erinaceus* (Choi et al. 2013), *Perenniporia fraxinea* (syn. *Fomitella fraxinea*) (Kim et al. 2008; Lee et al. 2002, 2006), *Pleurotus ferulae* (Choi et al. 2017), and *P. ostreatus* (Choi and Shin 1998). Among other screened species (*A. bisporus*, *H. erinaceus*, *L. edodes*, *P. cystidiosus*, *P. floridanus*, *P. pulmonarius*, and *P. salmoneostramineus*), the highest proteolytic activity (up to 54.28 U/mg) exhibited a crude extract of *L. edodes*. The molecular weight of the fibrinolytic enzyme was 50 kDa (Ali et al. 2017). The antiplatelet activity of aqueous extract of *L. rhinocerotis* was revealed using fresh human blood samples (Teo 2014). The molecular weight of the partially purified protease-like enzyme was 50–55 kDa. Fibrinolytic enzymes were also isolated from sclerotia of *L. rhinocerotis* as well (Ahmad et al. 2014).

The cultural liquid (CL) samples of eight polypore species *F. fomentarius*, *F. fraxinea*, *F. pinicola*, *L. sulphureus*, *T. gibbosa*, *T. hirsute*, *T. ochracea*, and *T. versicolor* have recently been screened for thrombolytic activity on human thrombus samples (Badalyan et al. 2016). The highest activity was detected in CL of *F. fraxinea* (up to 100%), *F. pinicola* (up to 85%), *F. fomentarius* (up to 83%), and *L. sulphureus* (up to 69%) strains, whereas the activity was weaker (20–55%) in *Trametes* species.

The screening of fibrinolytic activity of CL samples of two agaricoid (*F. velutipes* and *P. ostreatus*) and two polypore (*F. pinicola* and *G. lucidum*) mushrooms revealed the highest activity in *F. pinicola* (95%), followed by *G. lucidum* (55%), *P. ostreatus* (54.0), and *F. velutipes* (51%) (Badalyan et al. unpublished data). Milk-coagulating and fibrinolytic activities in *F. pinicola* and coprini species have also been detected (Badalyan et al. 2008b, 2016). It can be concluded that medicinal mushrooms may be considered alternative natural sources to develop novel pharmaceuticals with thrombolytic/fibrinolytic effects.

1.4.6 Neuroprotective and Antidepressant Effects

Inflammation, oxidative stress, mitochondrial dysfunction, and axonal transport deficits play a role in development many neurodegenerative disorders. Different neuroprotective strategies aim to limit these neurotoxic processes to delay disease progression. In spite of unsuccessful clinical trials, new therapies are emerging to ameliorating the process of traumatic and ischemic damage and to delay the progression of neurodegenerative processes.

Neurotrophins, such as nerve growth factor (NGF) and brain-derived neurotrophic factor (BDNF), are of great interest for survival, maintenance, and regeneration of specific neuronal populations in the adult brain. Several medicinal mushrooms, such as *Antrodia camphorata*, *G. lucidum*, and *H. erinaceus*, possess neurotrophic effects and could be considered as useful therapeutic agents in the management and/or treatment of depression and Alzheimer's, Huntington's, and Parkinson's diseases (Chen et al. 2006; Donatini 2011; De Silva et al. 2013; Lee et al. 2014b; Liu et al. 2015b; Muszynska et al. 2015; Phan et al. 2015, 2017a; Carhart-Harris et al. 2017; Wong et al. 2017; Rupcic et al. 2018).

The role of edible and medicinal mushrooms *A. bisporus*, *A. brasiliensis*, *C. militaris*, *G. lucidum*, *G. frondosa*, *H. erinaceus*, *L. edod*es, *L. rhinocerotis*, *O. sinensis*, *P. giganteus*, *T. versicolor*, *Termitomyces albuminosus*, and *T. fuciformis* in the treatment of neurodegenerative diseases and study of molecular mechanisms of neuroprotective and cognitive effects were recently reported (Wong et al. 2017; Tsuk et al. 2018). It was revealed that extracts and compounds obtained from mushrooms possess neuroprotective, and neuroregenerative effects for in vitro and in vivo assessments by targeting multiple physiological mechanisms and actions.

The neuroprotective effect has mainly been attributed to the discovery of various terpenoids that stimulate the production of NGF or BDNF. *Hericium* species are among the most praised edible medicinal mushrooms known as producers of bioactive compounds used for treatment of neurodegenerative diseases (Mori et al. 2009; Nagano et al. 2010; Donatini 2011; Lee et al. 2014b; Liu et al. 2015b; Samberkar et al. 2015; Thongbai et al. 2015; Brandalise et al. 2017; Chiu et al. 2018). Experimental studies have shown that polysaccharides and terpenoids (hericenones and erinacines) extracted from fruiting bodies and cultured mycelia of *H. erinaceus*, stimulate the synthesis of NGF, promote the growth and differentiation of neurons, and protect the cells against oxidative stress (Thongbai et al. 2015). The neuroprotective effects of *H. erinaceus* bio-products from homogenized, fresh fruiting bodies are manifested by attenuating the Aβ25–35-triggered damage of PC12 cells by significantly increasing cell viability and decreasing the release of lactate dehydrogenase (Liu et al. 2015b).

Hericenones and erinacines were also isolated from other *Hericium* species. NGF- and BDNF-induced corallocins and hericerin were detected in *H. coralloides* (Wittstein et al. 2016) and novel cyathane erinacines Z1 and Z2 with neurotrophin-inducing effects in mycelial cultures of *H. erinaceus* and *H. flagellum* (syn. *H. alpestre*) (Rupcic et al. 2018). Erinacine A-enriched mycelium of *H. erinaceus*

produced antidepressant-like effects by modulating BDNF/PI3K/Akt/GSK-3 signaling pathway in mice (Chiu et al. 2018).

The stimulation of neurite outgrowth by aqueous and ethanol extracts from sclerotia of *L. rhinocerotis* in rat pheochromocytoma adherent (PC-12Adh) cells at 20 μg/mL was revealed (Eik et al. 2012). Furthermore, the combination of 20 μg/mL aqueous extract and 30 ng/mL of NGF enhanced the neurite outgrowth by 42.1% compared to either aqueous extract (24.4%) or NGF (24.6%) alone. The neurite outgrowth activity was also revealed by using an aqueous extract of *L. rhinocerotis* mycelium and in combination with other natural products, such as *Ginkgo biloba* and curcumin (John et al. 2013).

The neurite outgrowth of aqueous extracts of sclerotium and mycelium of *L. rhinocerotis* in mouse neuroblastoma (Neuro-2a) cells was detected (Phan et al. 2013). At 20 μg/mL, the extract of sclerotium resulted in 38.1% of neurite-bearing cells, which was approximately twice the number of NGF-treated neurite-bearing cells. However, the aqueous extract of mycelium did not cause a significant increase in neurite outgrowth when compared to NGF treatment. An increase of neuritogenic activity of medicinal mushrooms in rat pheochromocytoma cells (Seow et al. 2013) and neuritogenic activity in PC-12 cell lines treated with *L. rhinocerotis* extracts (Seow et al. 2015) was also reported. The maximal neuritogenic activity in PC12 at 25 μg/mL of aqueous extract was 20.99% followed by ethanol extract (17.4%) and crude polysaccharides (16.4%). The hot aqueous extract (25 μg/mL) stimulated neuritogenesis similar to NGF (50 μg/mL). However, all extracts promoted neuritogenesis without stimulating the release of NGF by PC12 cells (Eik et al. 2012; Seow et al. 2015).

The aqueous extract obtained from sclerotia of *L. rhinocerotis* was also able to stimulate neurite outgrowth in dissociated cells of the brain, spinal cord, and retina of chick embryo (Samberkar et al. 2015). The 50 μg/mL extract induced maximal neurite outgrowth in the brain and spinal cord (20.8 and 24.7%, respectively), while this induction in retinal cells (20.8%) was achieved at 25 μg/mL.

1.4.7 Antiviral and Anti-allergic Effects

The prevention and treatment of viral infections remain a serious problem in modern medicine. Therefore, the search for effective antiviral medications, including those of natural origin, is in demand. Mushrooms are considered the producers of different bioactive compounds with antiviral effects; however, despite considerable work to reveal various features of mushrooms, the studies of antiviral effects are inadequate (Adotey et al. 2011; Santoyo et al. 2012; Teplyakova et al. 2012; Krupodorova et al. 2014; Teplyakova and Kosogova 2016; Doğan et al. 2018).

The antiviral effect of protein fractions isolated from mycelium of *L. rhinocerotis* against human papillomavirus (HPV) was revealed (Abdullah et al. 2013). The antiviral activity against dengue virus type-2 (DENV-2) strain was detected in the hot aqueous extract of *L. rhinocerotis* at IC_{50} of 520 μg/mL (Ellan et al. 2013). The extract also inhibited viral RNA synthesis by 99.7% but did not have a significant effect on DENV-2 virusicidal activity and viral attachment. The anti-dengue activity

correlated with carbohydrate content in the extract (2.0 mg/mL) suggesting that antiviral activity is attributed to polysaccharide content in the sclerotium.

The antiviral activities of extracts and polysaccharide fractions obtained from edible agaricoid mushrooms *B. edulis*, *L. edodes*, and *P. ostreatus* were detected against herpes simplex virus type 1 (HSV-1) (Santoyo et al. 2012). Aqueous extracts from mycelia of 11 polyporoid mushrooms showed antiviral activity against influenza A virus of birds A/chicken/Kurgan/05/2005 (H_5N_1) and humans A/Aichi/2/68 (H_3N_2) (Teplyakova et al. 2012). Among the tested species, *Daedaleopsis confragosa*, *Datronia mollis*, *Ischnoderma benzoinum*, *L. officinalis*, *Lenzites betulina*, *T. gibbosa*, and *T. versicolor* were suggested as perspective fungi to develop novel antiviral myco-medicines. It has been shown that polysaccharides, glycoproteins, melanins, nucleoside proteins, and terpenoids from several *Agaricomycetes* fungi exhibit antiviral effects against hepatitis, herpes, human immunodeficiency virus (HIV), influenza, West Nile viruses, as well as orthopox viruses, including the variola virus (Teplyakova and Kosogova 2016). Preparations from these mushrooms were also suggested for prevention of cancers with viral etiology.

The screening of antiviral activities of mycelia of ten basidiomycetous mushrooms against influenza A (serotype H_1N_1) and herpes simplex virus type 2 (HSV-2) showed that tested species inhibited the reproduction of influenza virus strain A/FM/1/47 (H_1N_1) in MDCK cells (Krupodorova et al. 2014). The species *Auriporia aurea*, *F. fomentarius*, *P. ostreatus*, and *T. versicolor* were also effective against HSV-2 strain BH in RK-13 cells, with similar levels of inhibition, for influenza virus. This is the original report of anti-influenza effects of *F. velutipes*, *Lyophyllum shimeji*, and *P. eryngii*. The anti-influenza and anti-herpes activities were also observed in *A. aurea*. The strain 353 of *T. versicolor* was found to be promising for the pharmaceutical industry as an anti-influenza and anti-herpes agent possessing low toxicity.

In a recent study, in vitro antiviral properties of *F. fomentarius*, *L. sulphureus*, *Morchella conica*, *M. esculenta*, *Ph. igniarius*, *P. ostreatus*, *Porodaedalea pini*, *Pyrofomes demidoffii*, *Terfezia boudieri*, and *Tricholoma anatolicum* were evaluated (Doğan et al. 2018). The results have demonstrated that the aqueous extract of *F. fomentarius*, *Ph. igniarius*, and *P. pini* showed considerable anti-HSV-1 activity, while methanolic and aqueous extracts of other species did not reveal any effects.

Recently the development of a mushroom-derived oral subunit vaccine from *Pleurotus* species has been suggested (Pérez-Martínez et al. 2015). However, antiviral bioactive compounds of fungal origin and mechanisms of their action remain subjects for further research.

Presently, allergies are considered an increasing problem worldwide. New strategies and a search for natural resources of anti-allergic compounds for preventive therapy are urgently needed. The anti-allergic potential of mushrooms has not been fully investigated yet. Using in vitro and animal assays, it has been shown that the influence of β-glucans on balance between Th1 and Th2 immune cells can result in an anti-allergic effect which has been described in *A. subrufescens*, *Armillaria ostoyae*, *F. velutipes*, *G. lucidum*, *G. tsugae*, *I. obliquus*, *Ph. linteus*, *P. ostreatus*, *P. pulmonarius*, *Tricholoma populinum*, and other *Agaricomycetes* fungi. However,

the bioactive compounds responsible for this effect have not been identified yet (Ellertsen and Hetland 2009; Bouike et al. 2011; Jesenak et al. 2014; Merdivan and Lindequist 2017).

The sesquiterpenes originally isolated from dichloromethane extracts of mycelium and fruiting bodies of honey mushroom *A. ostoyae* exhibited a significant degranulation-inhibiting effect on RBL-2H3 cells at non-cytotoxic concentrations (Merdivan et al. 2017). The purified compounds melleolides H and J significantly inhibited degranulation. Therefore *A. ostoyae* could be recommended as a natural preparation to support the treatment of allergies.

1.5 Nutritional and Dietary Values of Medicinal Mushrooms

Mushrooms are regarded as gourmet food with high nutritional and dietary values, as well as healthy DSs and myco-pharmaceuticals (Hobbs 2005; Khatun et al. 2012; Glamočlija et al. 2015; Kumar 2015; Sękara et al. 2015; Badalyan et al. 2016; Wu et al. 2016; Atila et al. 2017; Biswas et al. 2017; Gargano et al. 2017; Glamočlija and Soković 2017; Reis et al. 2017; Badalyan and Zambonelli 2019; Phan et al. 2018). Therefore, there is a significant potential to develop mushrooms as nutraceuticals and functional food for human wellness and their bioactive molecules for the production of drugs (Dutta 2013; Degreef et al. 2016; Süfer et al. 2016; Landi et al. 2017). The best implementation of mushroom products and myco-pharmaceuticals is to maintain a good quality of life (QoL), especially in immune-deficient and immune-suppressed patients; cancer patients receiving chemotherapy or radiotherapy; chronic blood-borne viral infections of hepatitis B, C, and D; different types of anemia; human acquired immunodeficiency syndrome (AIDS); herpes simplex virus (HSV); chronic gastritis and gastric ulcers caused by *H. pylori*; etc. (Cucuianu et al. 2004; El Dine et al. 2009; Wasser 2010, 2014, 2017; Adotey et al. 2011; Chang and Wasser 2012; Santoyo et al. 2012; De Silva et al. 2013; Thongbai et al. 2015).

The nutritional value and health benefits of mushrooms are determined by their chemical composition. Many reports on chemical composition and nutritional value of culinary-medicinal mushrooms belonging to different taxonomic and ecological groups were published (Badalian et al. 1997a; Badalyan and Rapior 1999; Hobbs 2005; Nunes et al. 2012; Palazzolo et al. 2012; Kalač 2013, 2016; Badalyan 2015, 2016; Da Silva et al. 2015; Adejumo et al. 2015; Teklit 2015; Thongbai et al. 2015; Badalyan 2016; Lalotra et al. 2016; Bandara et al. 2017; Kostic et al. 2017; Taofiq et al. 2017a). Mushrooms contain essential minerals, trace elements, vitamins, high levels of dietary fiber (chitin), proteins, unsaturated fatty acids, and nearly free of cholesterol and can be used for various human diets (Ayaz et al. 2011; Badalyan 2015, 2016; Lalotra et al. 2016; Landi et al. 2017; Phan et al. 2018). They are characterized by high contents of water and proteins and low energy value ranging from 50 to 70 kcal/100 g (Chang and Miles 2008; Dadakova et al. 2009; Chang and Wasser 2012; Cheung 2013; Kalač 2013; Glamočlija et al. 2015). Mushrooms are also regarded as a good source of vitamins (B_1, B_2, B_3, B_5, B_6, B_{12}, D, and H), phosphorus, iron, and ergosterol (Donnini et al. 2013; Zotti

et al. 2013; Kumar 2015; Kalač 2016). Macrofungi are considered a gourmet food because of their unique taste and flavor attributable to soluble sugars, eight-carbon derivatives, and various volatile organic compounds (Rapior et al. 1993, 1997; Breheret et al. 1998; Rapior et al. 2000; Nyegue et al. 2003; Badalyan and Zambonelli 2019; Kües et al. 2018).

Mushrooms contain 0.1–1 mg/g ergothioneine (2-mercaptohistidine trimethylbetaine), a water-soluble amino acid that naturally forms in some bacteria and non-yeast fungi. Mushrooms and meat are well-known sources of ergothioneine in human diet (Ey et al. 2007). The antioxidant effect of ergothioneine extracted from two *Pleurotus* species, i.e., *P. eryngii* and *P. citrinopileatus*, was recently reported (Yen et al. 2018).

The nutritive and medicinal value of ascomycetous mushroom, *M. esculenta*, is appreciated all over the world. Except for culinary properties, it has been used to treat gastric problems, to heal wounds, and reduce joint pain (Badalyan and Zambonelli 2019; Paul et al. 2018).

Based on their nutritional profile, mushrooms have been cultivated around the world as healthy dietary food (nutraceuticals) (Guillamón et al. 2010; Dutta 2013). The term "nutraceutical" is a conjunction of words "nutrition" and "pharmaceuticals"; therefore, nutraceutical is any substance which may be considered a food or part of the food that provides health-enhancing effects, including prevention and treatment of certain diseases (Barros et al. 2008; Farzana et al. 2017).

The nutritional and nutraceutical properties were evaluated in many mushroom species, e.g., *A. bisporus*, *P. ostreatus*, *P. sajor-caju*, and *S. commune* (Chang and Miles 2008; Chang and Wasser 2012, 2017; Chandrawanshi et al. 2017; Finimundy et al. 2018b; Ishara et al. 2018). The supplementation of food products with mushroom nutraceuticals (dietary fibers, proteins, and bioactive ingredients) obtained from *A. bisporus*, *Auricularia* spp., *B. edulis*, *F. velutipes*, and *L. edodes* be recommended to increase the quality and nutritional values of different types of products, such as dairy beverages, yogurts, bread, pasta, beer, etc. (Gregori 2014; Yen et al. 2015; Lu et al. 2016, 2018; Singh et al. 2016; Farzana et al. 2017; Heleno et al. 2017; Yuan et al. 2017; Francisco et al. 2018; Sknepnek et al. 2018). Mushroom-derived β-glucans possess effective health-enhancing effects and are ideal components for any baked goods, beverages, and DSs (Wasser and Didukh 2005; Zhu et al. 2015).

The potential of mushroom polysaccharides for the development of nutraceutical foods and drugs was recently summarized by Singdevsachan et al. (2016).

Among medicinal mushrooms, cultivated edible species *A. bisporus*, *A. campestris*, *A. auricula-judae*, *L. edodes*, *Pleurotus* spp., and *V. volvacea* are in high demand due to their nutritional value and pharmaceutical potential (Chang 1996; Chang and Buswell 1996; Chang and Miles 2008; Chang and Wasser 2012, 2017; Dutta 2013; Sękara et al. 2015). However, it should be noted that after consumption of *Auricularia* species, the Szechwan purpura syndrome may develop caused by the injury of thrombocytes (Hammerschmidt 1980; Giacomoni 2004; Brunelli 2009). Furthermore, the consumption of raw or undercooked *L. edodes* may cause shiitake dermatitis, a skin reaction caused by glucan lentinan (Nguyen et al. 2017).

The analysis of nutritional values (carbohydrate, crude fiber, protein, and fat contents), phenolic and flavonoid contents, and antioxidant capacity of wild mushrooms *Amanita calyptroderma*, *A. princeps*, *Astraeus odoratus*, *Heimiella retispora*, *Mycoamaranthus cambodgensis*, *Russula alboareolata*, *R. cyanoxantha*, *R. emetica*, *R. virescens*, and *T. clypeatus* and cultivated edible species *A. auricula-judae*, *Lentinus polychrous*, *L. squarrosulus*, *P. sajor-caju*, and *V. volvacea* showed that wild specimens contain higher content of fibers and bioactive compounds than cultivated specimens (Srikram and Supapvanich 2016).

The edible and medicinal mushrooms are also considered functional food or DSs ("nutriceuticals") due to their high contents of proteins, polyunsaturated fatty acids, α-tocopherol, phenols, vitamins, and other bioactive molecules (Chang 1996; Chang and Buswell 1996; Wasser and Didukh 2005; Chang and Miles 2008; Wasser 2010; Chang and Wasser 2012, 2017; Dutta 2013; Heo et al. 2014; Kumar 2015; Landi et al. 2017; Reis et al. 2017; Lu et al. 2018). Moreover, as a natural source of bioactive molecules, they may be used to develop myco-nutriceuticals and myco-pharmaceuticals.

The mushroom nutriceuticals can be extracted from fruiting bodies, mycelia, sclerotia, and spores' powder of basidiomycetous (*A. auricula-judae*, *C. cibarius*, *G. lucidum*, *L. edodes*, *Lentinus* (syn. *Pleurotus*) *tuber-regium*, *Lyophyllum decastes*, *P. eryngii*, *P. ostreatus*, *R. delica*, *S. commune*, *T. versicolor*, *T. fuciformis*, *V. volvacea*, and *Wolfiporia cocos*) and ascomycetous (*T. borchii* and *T. melanosporum*) macrofungi (Palazzolo et al. 2012; Adejumo et al. 2015; Teklit 2015; Venturella et al. 2016). The functional properties of uncooked and cooked ectomycorrhizal gasteroid fungi *Astraeus hygrometricus* have selective advantages and serve as a valuable raw material for the production of functional nutritional and pharmaceutical products with desired qualities (Pavithra et al. 2017).

A recent study addressed the chemical composition (glucans, glycerides, phospholipids, and sterols) and functional food value of cultivated culinary-medicinal mushroom *G. frondosa* possessing anti-inflammatory and antioxidant activities (Dissanayakea et al. 2018). Several compounds inhibited COX-1 and COX-2 enzymes at 100 μg/mL similar to NSAID, such as aspirin, ibuprofen, and naproxen.

The dietary supplements formulated from *Ganoderma* species are traditionally used in China, Japan, and Korea to improve life expectancy and prevent and treat many diseases, such as arthritis, asthma, bronchitis, CVD, gastritis, hypercholesterolemia, arterial hypertension, nephritis, neurasthenia, neoplasia, etc. (Paterson 2006; Cheng et al. 2010; Wang et al. 2012; Tan and Miyamoto 2015).

The screening of chemical composition, nutritional value, and bioactive properties of wild *Polyporus squamosus* mushroom revealed trehalose as the main free sugar and *p*-hydroxybenzoic acid as the main phenolic compound present in this species (Mocan et al. 2018). The highest levels of malic acid and β-tocopherol were detected in the fruiting bodies, while polyunsaturated fatty acids represented more than 57% of total fatty acids. As a source of polyunsaturated fatty acids, β-tocopherol, and bioactive compounds with antioxidant, antifungal, and antibacterial activities, *P. squamosus* can be further explored to develop new pharmaceuticals and DSs.

Several medicinal mushrooms (*A. campestris*, *A. auricula-judae*, *G. lucidum*, *O. sinensis*, and *Sanghuang porusbaumii*) have been used in traditional medicine for the prevention of diabetes (Zuomin et al. 1998). The usage of *G. lucidum*, *G. neojaponicum*, *G. frondosa*, *H. erinaceus*, *L. rhinocerotis*, *P. giganteus*, *T. camphoratus*, and many other mushroom products helps to reduce or prevent age-related neurodegenerative processes, such as Alzheimer's and Parkinson's diseases (Sabaratnam et al. 2013; Phan et al. 2015; Brandalise et al. 2017).

The genus *Pleurotus* (oyster mushrooms) comprises a group of mushrooms possessing high nutritional value and therapeutic properties. They are rich in proteins with essential amino acids, polysaccharides, dietary fibers, important minerals, and vitamins. The oyster mushrooms possess antiaging, anticancer, antimicrobial, anti-obesity, antioxidant, hypocholesterolemic, hypoglycemic, and hypotensive properties (Badalyan et al. 2008a; Baggio et al. 2010; Patel et al. 2012; Jayasuriya et al. 2015; Fu et al. 2016; Acharya et al. 2017; Debnath et al. 2017; Ebrahimi et al. 2017; Masri et al. 2017; Abidin et al. 2018; Adebayo et al. 2018; Finimundy et al. 2018a). They have been recommended as supplementary nutritional products for adjuvant therapy in patients with colorectal carcinoma. However, further clinical trials are needed to confirm the safety of described bioactive compounds of these mushrooms as alternatives to conventional drugs (Corrĕa et al. 2016; Finimundy et al. 2018a, b).

Fruiting bodies and mycelia of oyster mushrooms have been used in the human diet throughout the world due to their rich content of nutrients, including proteins, fibers, carbohydrates, minerals, vitamins, and lipids, and high culinary properties (Patel et al. 2012; Corrĕa et al. 2016). Several bioactive compounds have been identified in these mushrooms, including phenolics, polysaccharides, terpenes, and sterols (Corrĕa et al. 2016; Wu et al. 2016). Therefore, a high nutritional value and potent medicinal usage suggest that *Pleurotus* mushrooms can be considered renewable and easily accessible sources of innovative functional foods/nutraceuticals and pharmaceuticals with antioxidant, anti-inflammatory, antimicrobial, antitumor, and immunomodulatory effects (Gunde-Cimmerman 1999; Patel et al. 2012; Fernandes et al. 2015; Khatun et al. 2015; Corrĕa et al. 2016; Wu et al. 2016; Anjana and Savita 2017; Bello et al. 2017; Masri et al. 2017; Morris et al. 2017; Finimundy et al. 2018b). However, it should be noted that within 70 discovered *Pleurotus* species, only a few (*P. florida*, *P. ostreatus*, *P. pulmonarius*, and *P. sajor-caju*) are currently available in the market.

The nutraceutical and pharmaceutical potential of bioactive compounds isolated from *A. bisporus* (lectins), *A. auricula-judae*, and other *Auricularia* spp. (acidic polysaccharides), *G. frondosa* (grifolan, lectin), *Lentinus* (=*Pleurotus*) *sajor-caju* (lovastatin), and *O. sinensis* (cordycepin) (Xu et al. 2011, 2016a, b; Liu et al. 2015a; Prasad et al. 2015; Sękara et al. 2015) and general nutritional value, pharmacological properties, as well as potential for therapeutic applications of *L. edodes* (Finimundy et al. 2014; Thaper and Lakshmi 2017), *A. aegerita* (Landi et al. 2017), *Agaricus sylvaticus* (Monro 2003), and other mushrooms were also reported. Medicinal mushroom-based DSs produced from *A. brasiliensis*, *G. lucidum*, *G.*

frondosa, *L. edodes*, and *P. ostreatus* have already been approved for therapeutic use in Croatia (Jakopovich 2011).

Nutritional and nutraceutical properties of bioactive compounds (phenolics, flavonoids, antifungal proteins, immunostimulatory glucans, and lectins) derived from the wild edible mushroom *Lentinus squarrosulus*, as well as its antioxidant and wound-healing properties, were recently evaluated by Lau and Abdullah (2017). Both fruiting body and mycelium of *L. squarrosulus* were suggested as potential sources of functional ingredients. However, further clinical trials testing the mechanism of action of bioactive compounds in vivo and quality control of *L. squarrosulus* biomass obtained from different cultivation methods are recommended. The extracts obtained from fruiting bodies and sclerotia of medicinal mushrooms *P. tuber-regium* and *P. umbellatus* were suggested for the manufacturing of food supplements, pharmaceuticals, as well as cosmetic products and beverages (Badalyan et al. 2008a; Bandara et al. 2015).

Agaricomycetes coral mushrooms from genera *Ramaria* (order *Gomphales*) and *Clavaria* (order *Agaricales*) are widely distributed worldwide. However, most of these species have not been investigated for their nutritional and nutraceutical content. The information regarding the culinary status, biochemical and bioactive profiling of six *Ramaria* (*R. aurea*, *R. botrytis*, *R. flava*, *R. flavescens*, *R. rubripermanens*, and *R. stricta*) and six *Clavaria* (*C. amoena*, *C. coralloides*, *C. fragilis*, *C. purpurea*, *C. rosea*, and *C. vermicularis*) species collected in northwestern Himalayas was originally reported (Sharma and Gautam 2017). All the species were found to contain high levels of proteins, macro- and microminerals, carbohydrates, unsaturated fatty acids, essential amino acids, phenolics, tocopherols, anthocyanidins, and carotenoids. The studied mushrooms, particularly *R. botrytis* and *C. fragilis*, showed significant antioxidant and antibacterial activities against *E. coli*, *Klebsiella pneumoniae*, *P. aeruginosa*, *Streptococcus pneumoniae*, *Vibrio alginolyticus*, and *V. cholerae*. They are promising for future large-scale commercial exploitations and could be used as natural sources of antioxidant and antibacterial nutraceuticals.

Apart from basidiomes, the submerged mycelial culture of medicinal mushrooms represents a source to search for new safe and healthy myco-products with standardized quality. Currently, biotechnological cultivation of mycelia is progressing, and the production of mycelium-based biotech products from mushrooms is continuously improving (Asatiani et al. 2007; Chen et al. 2008; Elisashvili 2012; He et al. 2017a, b). In a recent study, the mycelium and cultural broth of *P. eryngii*, were reported as alternative sources of bioactive compounds with antioxidant, anti-inflammatory, and cytotoxic activities (Souilem et al. 2017). A protein-bound polysaccharide, nucleic acid, and amino acid content in mycelia and fruiting bodies of *F. velutipes* and *C. militaris* were analyzed, and the development of new products (amino acid drink or amino acid-containing food) from these mushrooms was suggested (Kim et al. 2014).

The prebiotics are food ingredients which stimulate the growth of beneficial microbiota. Oligosaccharides and fibers are major constituents of prebiotics. A recent trend in food science and technology has shown the efficacy of prebiotics to

modulate gut microbiota and attenuate several diseases, such as diabetes, obesity, and cancer.

Mushrooms are regarded as a potential source for the development of prebiotics containing different nondigestible polysaccharides, such as chitin, hemicellulose, α- and β-glucans, mannans, xylans, and galactans. From all mushroom polysaccharides, β-(1 → 3)-D-glucans and their peptide/protein derivatives (polysaccharide-peptide/protein complexes and proteoglycans) are essential prebiotics and possess immunomodulatory and antitumor effects by enhancing the growth of prebiotic bacteria in the gut (Bhakta and Kumar 2013; Singdevsachan et al. 2016). The medicinal mushrooms *C. versicolor*, *G. lucidum*, *G. frondosa*, *H. erinaceus*, *I. obliquus*, and *L. edodes* were reported as prebiotics due to the presence of immunomodulatory glucans regulating gut microbiota by inhibiting exogenous pathogens conferring health benefits to the host (Jayachandran et al. 2017; Diling et al. 2017). The prebiotic effect of Mexican *G. lucidum* (Meneses et al. 2016) and polysaccharides isolated from Polish wild mushroom (Nowak et al. 2017) have also been described.

Mushrooms have the effective capacity to absorb and accumulate trace elements from substrates, such as Selenium and other minerals. Thus, they could be used to produce mineral-enriched dietary food (Nunes et al. 2012; Milovanović et al. 2013, 2015a, b; Lalotra et al. 2016). The dietary Se has been recognized as an antioxidant, and the deficiency of this element is associated with numerous chronic degenerative diseases, various types of cancer, cardiomyopathies, and endemic osteoarthropathy (Rayman 2000; Combs 2001; Pedrero and Madrid 2009; Liu et al. 2016; Klimaszewska et al. 2017). High levels of Se (10–20 mg/kg dry matter) have been detected in widely consumable edible mushrooms, including *A. campestris*, *A. cesarea*, *A. macrosporus*, *A. silvaticus*, *Boletus aereus*, *B. aestivalis*, *B. appendiculus*, *B. edulis*, *B. erythropus*, *B. pinicola*, *G. lucidum*, and *Xerocomus badius* (Kalač and Svoboda 2000; Falandysz 2008). The effect of enrichment with Se on antioxidant, antifungal, and cytotoxic properties of mycelial extracts of *G. applanatum*, *G. lucidum*, *Lenzites betulina*, and *Trametes hirsuta* has recently been studied which revealed the stimulatory effect of Se on antioxidant activity of these species (Milovanović et al. 2015a, b).

The vitamin-enriched mushroom dietary food could also play an important role in the prevention of chronic diseases (Mehrotra et al. 2014). It has been shown that vitamin D_2-enriched mushroom (VDM) *A. bisporus* may provide a dietary source of vitamin D_2 and other bioactive molecules to prevent cognitive abnormalities associated with dementia (Bennett et al. 2013). However, this study supports the need for randomized clinical trials to determine whether VDM consumption can enhance cognitive performance in the wider population.

In recent years, mycosterols (MSs) have emerged as potential functional ingredients for the development of sterol-enriched food products and DSs (Heleno et al. 2017; Corrĕa et al. 2018; Francisco et al. 2018). The consumption of MS-enriched food may significantly reduce the levels of low-density lipoprotein, a major risk factor for CVD.

The isolation of novel bioactive MSs and study of their usage in formulations of nutritional products represent challenges for current nutritional and pharmacological research. The ergosterol (ergosta-5,7,22-trien-3β-ol) has been considered the principal sterol of hyphal membranes, followed by other derivatives, i.e., ergosta-5,8,22-trien-3-ol, ergosta-7,22-dien-3-ol, ergosta-5,7-dien-3-ol, and ergosta-7-en-3-ol (fungisterol) (Corrĕa et al. 2017).

Recent studies revealed new perspectives for application of mushroom-derived bioactive compounds in cosmetic industry (Taofiq et al. 2016a, b; Wu et al. 2016; Badalyan and Zambonelli 2019).

Presently, medicinal mushrooms in the Western Hemisphere are primarily used as DSs without a declaration of a medical indication. An overview of the principles of authorization, market access of herbal drugs in Europe, and the current status regarding mushrooms has been thoroughly described to support the development of legalized pharmaceutical preparations of mushroom origin in Europe (Lindequist 2013).

1.6 Current State of Epidemiological and Clinical Studies of Medicinal Mushrooms

Mushrooms have an established history of use in the traditional medicine of East Asian countries, Eastern Europe, as well as South and North America (Hobbs 1995; Stamets 2000; Wasser 2011; Guzmán 2015; Hapuarachchi et al. 2017; Gargano et al. 2017). Modern interdisciplinary research has validated the traditional knowledge which continues to demonstrate potent and unique therapeutic properties of bioactive compounds of mushroom origin.

Epidemiological and clinical studies of medicinal mushrooms conducted in Asian countries suggest that mushroom-derived natural products are used as part of different preventive and treatment strategies together with other therapies. They are mainly used as preventive agents against certain types of cancer, particularly gastrointestinal and breast cancers (Chang and Lee 2004; Sullivan et al. 2006; Bishop et al. 2015; Chaiyasut and Sivamaruthi 2017). Researchers are also focusing on the molecular mechanism behind anticancer action of mushroom-derived biomolecules (Joseph et al. 2017). Most of the available data comes from in vitro studies and in vivo experimental animal models. Therefore, systematic clinical trails should be initiated to translate the updated knowledge to clinical research.

Until now, the health-enhancing effects of various mushroom preparations in humans have been reported in more than 600 papers and reports. However, a few reviews address systematic clinical studies on the role of several medicinal mushrooms (e.g., *G. lucidum*, *G. frondosa*, *H. erinaceus*, *T. versicolor*, etc.) in the treatment of cancer, oncological immunological diseases, and diabetes (Smith et al. 2002a, b; Khatun et al. 2012, 2015; Hapuarachchi et al. 2016a, b; Wasser 2017; Rossi et al. 2018; Zmitrovich et al. 2019). Information on different clinical trials and epidemiological studies of medicinal mushrooms can also be found in several books (Hobbs 1995; Stamets 2000; Powell 2014; Rossi et al. 2018).

Epidemiological reports on the dietary consumption of mushrooms to promote health and prevent different illnesses mainly come from Brazil, China, India, Japan, and Korea. They have indicated that consumption of medicinal mushrooms in the form of fruiting bodies or extracts may prevent or delay the development of cancer. Previously reported epidemiological studies of gastric and breast cancers were mainly carried out with consumption of *G. frondosa*, *G. lucidum*, *F. velutipes*, *Hypsizygus marmoreus*, *Pholiota nameko* (Ikekawa 2001, 2005; Hara et al. 2003; Zhang et al. 2009; Shin et al. 2010; Rossi et al. 2018), *A. bisporus*, *L. edodes*, and other edible medicinal mushrooms in combination with green tea (Zhang et al. 2009). The results have shown that three times daily consumption of 3–5 g of hot water extract of *A. subrufescens* improves quality of life (QoL) by strengthening the innate healing power and immune system (Liu et al. 2008). The preclinical assessment of safety of the shiitake (*L. edodes*) on biochemical, haematological, and antioxidant parameters in rats showed that the daily intake of 100 mg/kg of mushroom could have potential health-enhancing effects (Grotto et al. 2016).

Currently, more than 130 medicinal properties (analgesic, antibacterial, antifungal, anti-obesity, antioxidant, antiparasitic, antitumor, antiviral, cardioprotective, immunomodulatory, neuroprotective, anti-allergic, etc.) were described in different mushrooms, such as *A. bisporus*, *A. blazei*, *A. cinnamomea*, *G. lucidum*, *G. frondosa*, *H. erinaceus*, *L. edodes*, *P. cornucopiae*, *P. eryngii*, *P. ostreatus*, and *T. versicolor*.

Modern clinical practices in China, Japan, Korea, Russia, and several other countries rely on medicinal mushroom-derived drugs (MMDD) and DSs to increase the tolerance to cancer therapy in cancer patients (Wasser 2010, 2017; Suzuki et al. 2013; Tangen et al. 2015; Twardowski et al. 2015; Del Buono et al. 2016; Tanaka et al. 2016; Tsai et al. 2016; Wasser 2017; Zmitrovich et al. 2019). In many clinical studies, mushrooms were used as an adjuvant treatment with conventional chemo- or radiotherapy for different types of cancer (bladder, brain, breast, liver, lung, ovary, prostate, stomach, etc.) to reduce the side effects of cancer treatment (e.g., hair loss, nausea, and loss of appetite) (Smith et al. 2002b; Sullivan et al. 2006; Suárez-Arroyo et al. 2017; Rossi et al. 2018). However, no evidence revealed the proper dosages and treatment duration for many species, sometimes due to poor study design, non-standardized mushroom preparations, statistical methods, etc. (Wasser 2017).

Currently, both edible and inedible mushrooms used in traditional medicine are undergoing clinical trials (Figs. 1.1 and 1.2). The Asian traditional medicine has proved the significance of inedible reishi (*G. lucidum*) and edible shiitake (*L. edodes*) mushrooms. Other well-known medicinal species, such as *Coriolus* (syn. *Trametes*) *versicolor*, *F. fomentarius*, *Fomitopsis officinalis*, *I. obliquus* (Chaga), *Ph. linteus*, and *P. betulinus*, are also used for the treatment of gastrointestinal disorders, various forms of cancer, bronchial asthma, etc. (Grienke et al. 2014; Chen et al. 2016).

Preclinical and clinical studies of *G. lucidum* polysaccharide (GLPS), as an approved non-hormonal drug, have been used for treating refractory myopathy and other diseases (Zeng et al. 2018). As herbal products, *A. blazei*, *Cordyceps* species,

C. versicolor, *G. lucidum*, *L. edodes*, and *T. camphoratus* are widely used in TCM together with pomegranate, green tea, garlic, turmeric, and *Artemisia* plants (Lee et al. 2012).

Although in vitro and animal studies as well as the history of usage of mushrooms in TCM confirm the ethno-mycomedicinal knowledge, there is no scientific evidence to support their efficacy for treatment of human diseases. Therefore, consumers should critically evaluate information about the medicinal properties of mushrooms (Money 2016; Mustonen et al. 2018). Thus, systematic experimental, epidemiological, and clinical studies are required to fully explore the pharmacological potential of mushrooms and develop mushroom pharmacopeia (Taofiq et al. 2016a, 2017a; Wasser 2017).

Currently, a large number of clinical trials are carried out using edible and inedible wild and cultivated mushrooms, including *A. subrufescens*, *G. lucidum*, *G. frondosa*, *H. erinaceus*, *I. obliquus*, *L. edodes*, *O. sinensis*, *Ph. linteus*, *S. commune*, and *T. versicolor* (Figs. 1.1 and 1.2). The anticancer, immunomodulatory, hypoglycemic, and hypolipidemic activities of mushrooms were mainly investigated (Xiao et al. 2004; Hobbs 2005; Oba et al. 2007, 2009; Standish et al. 2008; Eliza et al. 2012; Ren et al. 2012; Aleem 2013; Suzuki et al. 2013; Donatini 2014; Guggenheim et al. 2014; Fritz et al. 2015; Motoi et al. 2015; Del Buono et al. 2016; Frost 2016; Wasser 2017).

Different extracts obtained from fruiting bodies, mycelial biomass, spores, and pure β-glucans (e.g., lentinan from *L. edodes*, schizophyllan from culture broth of *S. commune*, PSK or PSP complex from *T. versicolor*) have been tested to treat oncological diseases in clinical trials (Ren et al. 2012; Roupas et al. 2012; Suzuki et al. 2013; Donatini 2014; Del Buono et al. 2016; Ina et al. 2016; Wasser 2017). Clinical trials, and epidemiological data evaluate the efficacy and safety of mushroom-derived β-glucans, mainly lentinan, Maitake D-fraction, and schizophyllan in the prevention and treatment of cancer (Aleem 2013; Wasser 2017). Future randomized controlled trials to reveal the clinical efficacy of mycotherapy and the mechanisms of action of mushroom-derived anticancer compounds on long-term survival, tumor response, host immune functions, inflammation, and QoL in cancer patients should also be addressed (Rossi et al. 2018).

Some clinical studies have elucidated the mechanisms responsible for the immunomodulatory effects of β-D-glucans. They have shown to activate cytotoxic macrophages, monocytes, neutrophils, NK cells, cytokines (interleukins and interferons), and colony-stimulating factors (CSF) that trigger complementary and acute phase responses (Johnson et al. 2009; Choi et al. 2014; Dai et al. 2015; Kang et al. 2015; Tanaka et al. 2016).

Contemporary studies directed to biological activities of fungal β-glucans have gained significant attention in biomedical science. However, the mechanism of biological activity of fungal β-glucans has not been completely understood. The physicochemical and biological characteristics of β-glucans isolated from different agaricoid and polyporoid edible and medicinal mushrooms have recently been discussed by Zhu et al. (2015). The polysaccharide fraction obtained from *A. bisporus*, consisting of 90% β-glucans, induced the production of the tumor necrosis factor

alfa (TNF-α) in vitro. However, administered in vivo, β-glucans have lost their ability to stimulate the immune response as observed using an in vitro animal model (Volman et al. 2010).

The recent results of clinical studies involving *A. subrufescens*, *G. lucidum*, and *I. obliquus* mushrooms revealed the effects of β-glucans on cancer cells (Frost 2016).

It has been shown that the immunomodulatory effects of β-glucans are used for prevention of viral infections (Vetvicka and Vetvickova 2015). The feeding of influenza virus-infected mice for 2 weeks with a mixture of glucans obtained from fruiting bodies of *G. frondosa* and mycelia of *A. brasiliensis*, *I. obliquus*, and *L. edodes* significantly reduced the clinical symptoms of infection. These results suggest that the consumption of dietary glucan might be useful as a complementary or alternative approach for the treatment of influenza infection. Further clinical application of β-glucans and studies of structure-activity relationship to develop β-glucan-enriched functional food products should be performed.

The five major macrofungi *A. blazei*, *G. lucidum*, *G. frondosa*, *O. sinensis*, and *T. versicolor* used in the treatment of oncological diseases appear to improve the effects of chemotherapy (Guggenheim et al. 2014).

T. versicolor-derived PSK is widely used in Japan as an adjuvant immunotherapeutic agent to treat a variety of cancers, including lung cancer. The meta-analysis of *T. versicolor* effects demonstrated an increased rate of survival, particularly in breast, gastric, and colorectal cancer patients (Eliza et al. 2012). Fifteen of 17 preclinical studies supported the anticancer effects of PSK through immunomodulation and tumor-inhibiting effects in vivo which resulted in reduced tumor growth (Fritz et al. 2015). The usage of *T. versicolor* as part of immune therapy for breast cancer treatment showed that polysaccharide constituents of the mushroom as concurrent adjuvant cancer therapy might be warranted as part of comprehensive cancer treatment and secondary prevention (Standish et al. 2008). Eight randomized controlled clinical trials, including 8009 patients with gastric cancer treated with PSK in combination with chemotherapy, indicate that adjuvant immune-chemotherapy improved the survival of cancer patients after gastric resection in comparison to chemotherapy alone (Oba et al. 2007).

The purified β-glucan lentinan from *L. edodes* is a well-established drug used for the combined treatment of oncological diseases in Japan. Moreover, preparations for intravenous and per os administrations of lentinan are currently available. Five studies, involving 650 patients with non-resectable or recurrent stomach tumors, demonstrated that patients treated with chemotherapy and lentinan had a significantly higher survival rate compared to patients treated only with chemotherapy (Oba et al. 2009).

A randomized, double-blind, placebo-controlled trail in children with recurrent respiratory tract infections showed that treatment with pleuran, a β-glucan isolated from *P. ostreatus*, reduced symptoms of atopy related to these infections (Jesenak et al. 2014).

A few in vivo studies have been performed with *Ganoderma* species (Jin et al. 2012; Hapuarachchi et al. 2016a, b, 2017; Klupp et al. 2016; Wasser 2017). In many cases, it has been questioned whether *Ganoderma* is solely a DS for well-being or merely a helpful "medication" for regenerative purposes. No conclusive report is

available on clinical trials using *Ganoderma* species as a direct control agent for the diseases. In addition, there is no evidence supporting the usage of *Ganoderma* as a potential supplement for oncological or other diseases since no preclinical trials have been performed to date. The purified substances isolated from *Ganoderma* have been investigated to reveal the molecular mechanisms responsible for antitumor and immunomodulatory effects. It was concluded that the standardization and quality control of *Ganoderma* species, cultivation processes, and commercial formulations are required to consider these species as natural products for the prevention and treatment of diseases (Hapuarachchi et al. 2017). Further experimental, epidemiological, and clinical studies should be carried out to identify molecular targets and investigate the association between *Ganoderma* intake and disease risk. Moreover, the efficacy, dosage, and safety, alone or in combination with chemotherapy or radiation treatment, should also be investigated (Hapuarachchi et al. 2016a, b).

A review of more than 250 clinical studies of *G. lucidum* and other *Ganoderma* species was published by Jin et al. (2012). The patients who received *G. lucidum* in addition to chemotherapy and radiation treatment revealed a stronger response to conventional treatment compared to patients with conventional treatment alone. Moreover, this combination treatment also improved their QoL. It was concluded that *G. lucidum* could be administered as an adjunct to conventional treatment considering its potential of stimulating the immune system of the host.

Reishi mushroom *G. lucidum* is also frequently used for the treatment of CVD (Klupp et al. 2015). Five medical studies comparing *G. lucidum* with placebo in 398 patients with type 2 diabetes showed that the daily usage of *G. lucidum* or a mixture of *G. lucidum* (75%) and *O. sinensis* (25%) does not support the use of *G. lucidum* for the correction of cardiovascular risk factors in patients with type 2 diabetes mellitus. Future research of *G. lucidum* should be placebo-controlled and adhere to clinical trial-reporting standards. The results of a double-blind, randomized, placebo-controlled trial found no significant effect of administration of 3 g/day *G. lucidum* or a combination of *G. lucidum* with *O. sinensis* for 16 weeks on HbA1c and fasting plasma glucose levels of a small number of patients with type 2 diabetes (Klupp et al. 2016).

Many ex vivo and in vivo studies have revealed antidiabetic, anticancer, anti-inflammatory, antimicrobial, antimutagenic, antioxidant, antiparasitic, hepatoprotective, immunomodulatory, and other pharmacological activities in the medicinal mushroom *A. blazei* Murrill sensu Heinemann (syn. *A. subrufescens*), but only 17 clinical studies and 2 case reports on *A. blazei* were found (Therkelsen et al. 2016). The nutritional and therapeutic properties of *A. blazei*, with emphasis on its chemical composition, as well as clinical trials were recently reviewed (Da Silva de Souza et al. 2017).

The findings of clinical trials involving the regular consumption of *A. brasiliensis* Wasser (syn. *A. blazei* Murrill sensu Heinemann) (Ahn et al. 2004; Talcott et al. 2007; Motoi et al. 2015) and *G. frondosa* (Kodama et al. 2002; Konno 2009; Rajamahanty et al. 2009) by healthy volunteers showed improved immunity and, in

single cases, increased rate of survival and general quality of live (QoL). However, a randomized, double-blind, placebo-controlled trial showed that the extract of *A. blazei* had no stimulating effect on IL-6, IFN-γ, or TNF-α levels in elderly females (Lima et al. 2011). Further clinical trials, using reliable statistical methods and standardized preparations, are needed to establish the efficacy of *A. blazei* as a therapeutic agent (Therkelsen et al. 2016).

The results of a randomized, double-blind, placebo-controlled clinical trials suggest that the dietary supplementation with *A. sylvaticus* improves the haematological and immunological parameters and reduces blood glucose levels in colorectal cancer patients (Fortes et al. 2008, 2009).

The medicinal properties (anti-asthmatic, anticancer/antitumor/immunomodulatory, anticoagulant, anti-inflammatory, antimicrobial, anti-obesity, antioxidant, antiviral, and neuroprotective) of *L. rhinocerotis* were recently reviewed (Nallathamby et al. 2018). In vitro results suggest that fruiting bodies and sclerotia of this species can be considered as alternative therapeutic strategies in the management of non-communicable diseases. Therefore, additional studies including in vivo clinical trials are needed to scientifically validate the usage of mycochemicals of *L. rhinocerotis* and develop therapeutic products.

Numerous bioactive compounds derived from medicinal mushroom *H. erinaceus* (Yamabushitake) have been developed into food supplements and alternative medicines (Thongbai et al. 2015). A double-blind, parallel-group, placebo-controlled trial was performed on Japanese men and women diagnosed with mild cognitive impairment to highlight the efficacy of oral administration of *H. erinaceus* (250 mg tablets containing 96% of *H. erinaceus* powder three times a day for 16 weeks). The obtained results have shown that *H. erinaceus* is effective in improving mild cognitive impairment (Mori et al. 2009).

A prospective randomized trail of 25 patients comparing the efficacy of *H. erinaceus* versus essential oils against *H. pylori* infection revealed that patients who received *H. erinaceus* showed negative results for Pyloritop® test in 89.5% of cases in contrast to 33.3% for patients who started with essential oils (Donatini 2014). It was suggested that *H. erinaceus* could be considered an alternative to antibiotic therapy against *H. pylori*-associated diseases. Additional randomized studies versus referential therapy should be performed to focus on treatment without adverse effects.

In a double-blind, placebo-controlled clinical study, oral administration of *H. erinaceus* (250 mg tablets with 96% mushroom powder, three times a day, for 16 weeks) improved cognitive abilities of 50–80-year-old patients (Mori et al. 2009). Another clinical study revealed the antidepressant effect of *H. erinaceus* (Nagano et al. 2010) which may be due to its high 5-hydroxytryptamine content (Muszynska et al. 2015).

A previous clinical study of oyster mushroom has revealed a significant reduction of blood glucose, arterial pressure, and cholesterol levels of diabetic patients without any deleterious effects on the liver and kidney (Khatun et al. 2007). However, well-designed randomized controlled trials with long-term consumption

of this mushroom are needed to prove the bioactivity and safety of the product for diabetic patients.

Proceeding from the results of epidemiological and clinical studies of medicinal mushrooms, it should be noted that high-quality, long-term, randomized, double-blind, placebo-controlled clinical studies should be continued in the future.

1.7 Conclusions and Future Prospects

Mushrooms can be used as nutritional products, tonics, medicines, cosmetics, and natural biocontrol agents. The multidimensional nature of the global mushroom-cultivation industry, its role in addressing critical issues faced by humankind, and its positive contributions are noteworthy. Furthermore, mushrooms can serve as agents for promoting economic growth in a society. Mushrooms produce chemically diverse compounds with a broad spectrum of biological activities. In vitro assays, animal studies, and clinical trials justify the traditional experience and suggest a great potential of mushroom-derived compounds and products for the prevention and treatment of various diseases. In view of promising results, more efforts are needed to explore the therapeutic potential of medicinal mushrooms and promote the development of drugs. The important tasks include the realization of clinical studies, development of high-quality mushroom-derived products with standardized procedures, and their sustainable production under controlled conditions.

The present review covers the potential benefits of medicinal mushrooms related to nutrition and diet, functional food supplements and bioactive ingredients for prevention of diseases, and reduction of side effects in patients receiving conventional therapies. The complementary medicine is becoming one of the options that patients use to address their distress during and after treatment. Limited information exists about the costs and benefits of adjunct non-pharmacological treatments using medicinal mushrooms. Further research is needed in this field to expand knowledge preclinical and clinical investigations based on relevant data and facilitate the comprehensive assessment of therapeutic value of mushrooms.

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